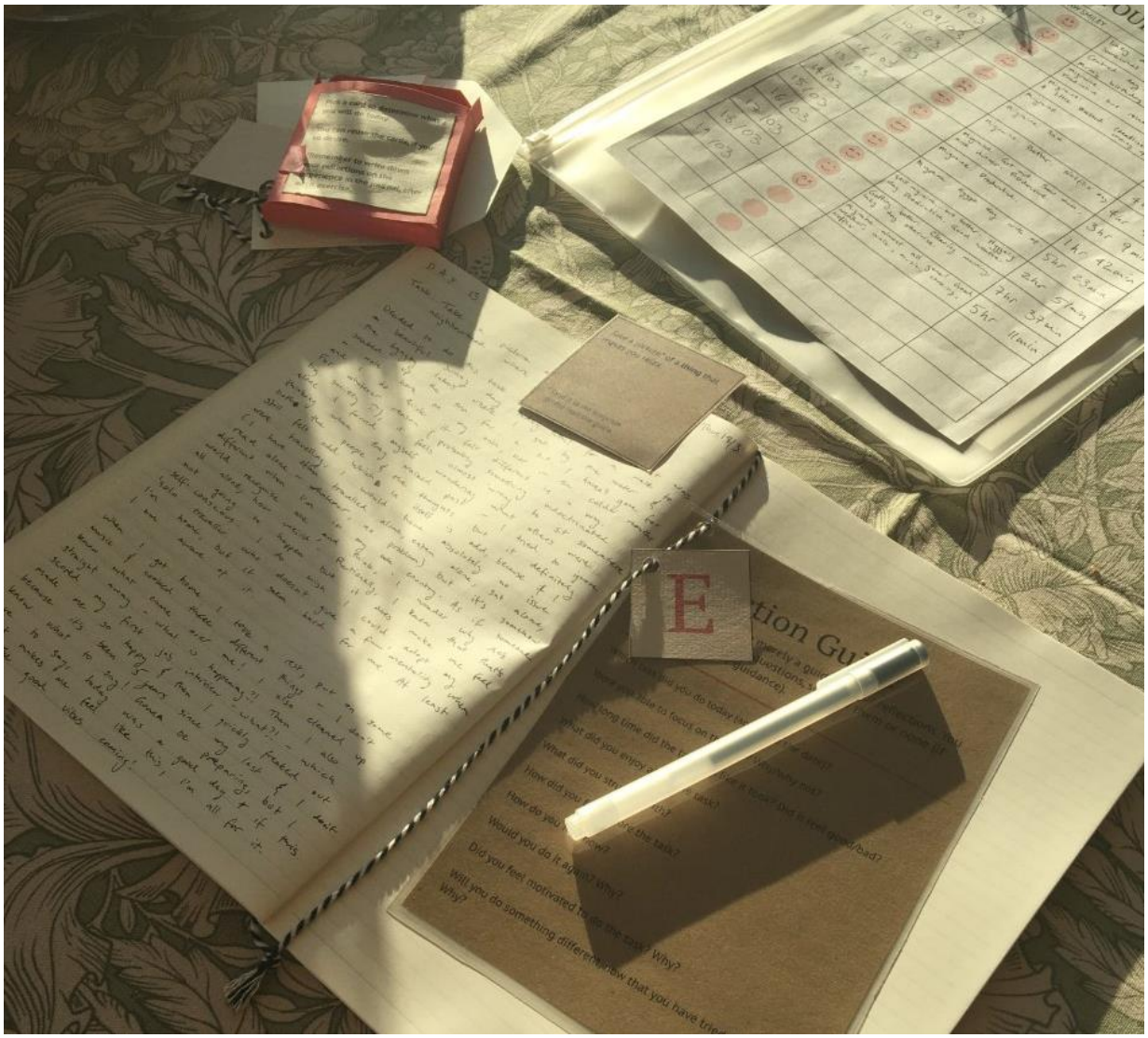
Master Thesis 2021

Designing towards Self-appreciation Routines for Young Women



STADS Code: KISPECI1SE

Anne Katrine Svejgaard Barrett
kaah@itu.dk

Emily Howkins Lange
emhl@itu.dk

Abstract

In this report, we will explore issues of self-esteem in young women. We will use a Research through Design approach within design research, with a focus on interaction design, to explore and understand the problem. We will further look into the extent to which Social Media has an effect on self-esteem in young women. In order to gain insights into how young women experience self-esteem and what can help mitigate feelings of low self-esteem, we conducted a cultural probe study over the course of 4 weeks, with 5 young women. This was followed by a participatory workshop, in which the users could reflect over their experience with the cultural probe kit and give further insights to the dynamics that come into play when dealing with feelings of low self-esteem. We then coded and analysed our data from the cultural probe and our workshop. These findings were collated into a set of 8 design arguments. Moreover, we applied our design arguments in a design workbook consisting of 5 design proposals. This served as a demonstration of how we could apply our arguments, and they can work as a theoretical design contribution.

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Introduction

This thesis project started with a concerned interest in the growing number of young Danes experiencing depression, anxiety and suicidal ideation (Sundhedsstyrelsen, 2017). We were interested in why and how this was happening. We began to examine and attempt to understand the threats and causes of these states of mind. It became clear that it was especially low self-esteem over longer periods that was a common denominator in the creation of more serious mental conditions. Experiences of low self-esteem and low self-worth are often associated with a growing tendency of needing to perform more and better in social contexts, in school and academia, at work and on social media. To delve further into this problem, we began to ask ourselves questions that would either refute or confirm our assumptions on this issue. These questions included: What affects young adult's self-confidence and self-esteem? What are methods for improving self-esteem? What digital tools exist in helping young adults address these feelings, and do they actually work?

We initially found that social media usage and social pressures to overperform might be part of the issue. As aspiring digital designers, we naturally found an interest and a feeling of responsibility to examine whether specific design features could be a causing factor in experiencing low self-esteem. Further, we wanted to explore how a digital artifact could be designed with more ethical responsibility towards the user's intended and unintended emotional experience.

In order to understand our research context, we wanted to investigate the problem. To do this we looked into who experiences low self-esteem and what long-term consequences they might experience. Furthermore, it seemed relevant to understand the inner workings of self-esteem - how it is perceived and defined. Finally, our aim was to identify major triggers of low self-esteem and in turn what can help mitigate those feelings. This will be explored in our problem analysis and related works. The exploration into this problem worked as the foundation for our research question:

> **How can we assist young women in practicing self-appreciation through a digital artifact, and how can this improve their self-esteem and happiness long-term?**

In attempting to answer our research question, we explore the qualities and dynamics of self-esteem in young women. This is with the intention of creating a design artifact that can assist in practicing self-appreciation, as a means of addressing feelings of low self-esteem. We did this by applying methods grounded in design research. Our primary method for gathering empirical data was the development and use of a cultural probe deployed with five young women based in Denmark for four weeks. This was followed by a participatory workshop to evaluate their experiences with the designed features within the cultural probe. Finally, this led to a collection of design arguments grounded in our empirical data. These design arguments are supported by a design workbook where these are applied.

Investigating the problem

In this chapter we will investigate the problem of young women experiencing low self-esteem. We will be defining the inner workings of self-esteem and seek to understand the causes of long term low self-esteem. Further, we will look into consequences and mitigations regarding low self-esteem. In order to seek an understanding of how digital artifacts might have an impact on this problem, we will also be looking at the usage of smartphones and Social Media (SoMe) through an analysis of their three interaction forms. Finally, we will approach our design space by exploring related works in this research context.

What is self-esteem?

In this section, we will look further into the definition of self-esteem. This will also cover a description of the extent to which low self-esteem is a growing issue because there is a rising number of people experiencing the long-term consequences of low self-esteem. These are mental health issues such as depression, anxiety, and stress (Wilbum & Smith, 2005; Emler, 2001).

In 2017, the Danish health department published a report on children and youths who experience anxiety or depression, ADHD, and/or eating disorders (Sundhedsstyrelsen, 2017). This is based on data from 2006 to 2016 with children and youths in the ages from 0-18. The report shows a rise in the diagnosis of anxiety and depression in individuals, from 2354 to 7189 over the course of 10 years. This is a rise of 206,57% in children and youths. In 2016, 92,9% of individuals diagnosed with anxiety and depression were in the ages of 12-18. We find this to be a major argument in the importance of looking into proactive ways to reduce this rise of anxiety and depression.

Self-esteem, self-confidence, self-efficacy, self-worth – there are an array of words in which to describe this emotional phenomenon. Self-esteem is believed to be a vital part of mental health. At the same time, talking about low self-esteem is believed to be somewhat taboo, as it can be perceived as an indicator of lack of success (encyclopedia.org, n.d.).

Self-esteem is defined as the evaluative component of the self-concept (Rosenberg et al., 1995). The self-concept is the self-evaluation of being either likable or unlikeable. This self-evaluation can be based on personal beliefs about skills, abilities and future outcomes. Further, there is a cognitive and affective component to self-esteem, and here it is believed that the affective component outweighs the cognitive and rational component of the self-concept. An example here is people who dislike themselves in spite of evidence showing that they have reasons to feel good about themselves. (Rosenberg et al., 1995)

Self-esteem can be conceptualised as global and multidimensional (Rosenberg et al., 1995). Global self-esteem is best understood as a hierarchical setup with three elements: performance, social and physical self-esteem. Each element can be further broken down. Performance self-esteem deals with one's general competence. This includes intellect, agency, efficacy, self-regulation, self-confidence, and the like. Social self-esteem is concerned with how individuals believe that they are perceived by others. Here, it is perception rather than reality that is of importance. People with low social self-esteem are believed to experience more social anxiety than others. Finally, physical self-esteem is concerned with bodily performance as well as how the body looks. This can also be affected by physical stigmas or race and ethnicity (ibid.).

For measuring global self-esteem, the most widely used scale is the Rosenberg scale (1965). This consists of ten statements such as:

- On the whole, I am satisfied with myself
- I certainly feel useless at times
- I take a positive attitude toward myself

Whether self-esteem is a stable personality trait or a dynamic and context-specific state is still up for debate. In this project, we are working with self-esteem as something inherent in all individuals, but as a dynamic state that can be affected in a positive or negative direction. Upkeeping self-esteem is here regarded as a highly habitual practice.

If we look at other research papers on the subject, we see that low self-esteem is

considered a normal emotional state, to the extent where it can have long term consequences for individuals. In an article from 2005, Wilbum and Smith examined the connections between long-term low self-esteem and stress leading to suicidal ideation in young adolescents. The research was carried out as a reaction to rising suicide levels in the United States. A major finding in this article was that negative stress and self-esteem significantly predicted suicidal ideation. Here, negative stress for adolescents is linked to the life changes that can occur in this period of an individual's life. Wilbum and Smith (2005) give examples such as academic pressure, work-related problems, interpersonal difficulties, and loss of important relationships. Experiencing major stressors in life is one of the key ingredients when concocting a nice cocktail of low self-esteem.

Emler (2001) describes failure and successes as factors that have an impact on low self-esteem. The way they impact self-esteem is very subjective to each individual. The objective positive or negative experiences we have in life, do not necessarily dictate whether we experience high or low self-esteem; the matter is more complex in nature. Rather, it is the successes or failures in areas that the individual finds most significant to themselves that they will lay their foundation of self-esteem upon. For some, high performance in academia might be what will affect their self-esteem, where others might be more affected by their performance on social media. Furthermore, Emler (2001) suggests that successes and failures are experienced through the individual's own construct of the story that they will remember. This construction of a story is a learned ability where individuals who experience more low self-esteem, will find it more difficult to frame positive experiences as successes. Emler (2001) describes this as such:

> *The greater puzzle perhaps is why some people will discount good performances to protect a view of themselves as worthless? Yet it does seem that the nature of these biases reflects self-esteem. Those blessed with high self-esteem ignore all evidence of inadequacies. Those who lack esteem equally consistently deny that there is any positive evidence, using many of the same tactics in reverse.* (Emler, 2001, p. 40)

Once again, we see evidence that self-esteem is built on one's own self-narrative, which is based on a very habitual pattern of thought and reasoning as to how one performs.

Another element that affects self-esteem is experiences of rejections and acceptance. Emler (2001) argues that rejection is mainly linked to areas of work and relationships with others. It is particularly in young adolescents that acceptance becomes important in building self-esteem.

Acceptance of appearance is also described as an important aspect of self-esteem. Especially in women, who will seek acceptance based on their appearance, whether it be through the eyes of their significant others, the public or social media. This is not a new phenomenon. Cooley described in 1983, the need for acceptance from others and the imagined judgement from others in his essay 'Looking Glass-self'. The looking glass-self is a metaphor for how individuals will use an imagined person's judgement to pass judgement over their own appearance, manners, aims, deeds, character, friends, and so on. There are three components to this: the imagined appearance to others, the imagination of others' judgement and the feeling of pride or mortification based on this judgement. What has changed in regards to this phenomenon is the way in which individuals can seek acceptance on SoMe. SoMe allows for quick and easy feedback on visual and textual materials from people all around the world. The individual is, in other terms, not only being judged by an inner imagined other, but real life strangers mediated through SoMe.

This leads us to what Emler (2001) describes as the most significant influence on self-esteem: parents and significant others. It is through the acceptance, love and self-reflection in significant others that the foundation of self-esteem is built. First, is the foundation for self-esteem, which is brought forth in a parents' ability in making their children feel loved, capable and worthy. In turn, if the opposite is true, if children feel unloved, incapable and unworthy, it can result in low self-esteem that is almost immutable. Secondly, individuals seek to build on their self-esteem through other relationships in life that they believe are significant to them. In this project, we will not be looking further into what constitutes good parenting or how parents should raise their children. We simply know that there is a relationship between an individual's

upbringing and their self-esteem. We will, on the other hand, be looking further into how the individual can work on their self-perception, and how this in turn affect their relationships with significant others.

We see that significant others are at the centre of previously mentioned causes for low self-esteem, as it is through their (imagined) eyes that individuals will pass judgement onto themselves. We also see a tendency in how individuals seek positive judgement from others through SoMe. We might even suggest a clear correlation between higher levels of depression and anxiety, as a result of low self-esteem, and the use of SoMe in young adults.

In the next section we will continue to explore the technical phenomenon of SoMe, as a digital platform for mass communication. We will be looking into the growth of SoMe, how design of SoMe can affect its users and the interactive forms of SoMe and the smartphones that it is often used on.

Social Media as a growing problem

SoMe really started gaining traction in 2004 with the SoMe platform MySpace (Ortiz-Ospina, 2019). Since 2004, there has been a significant rise in SoMe usage. Facebook has gone from zero to over 2 billion users since 2008 to 2018, and Instagram from 117 million to 1 billion from 2013 to 2018. (Figure 1)

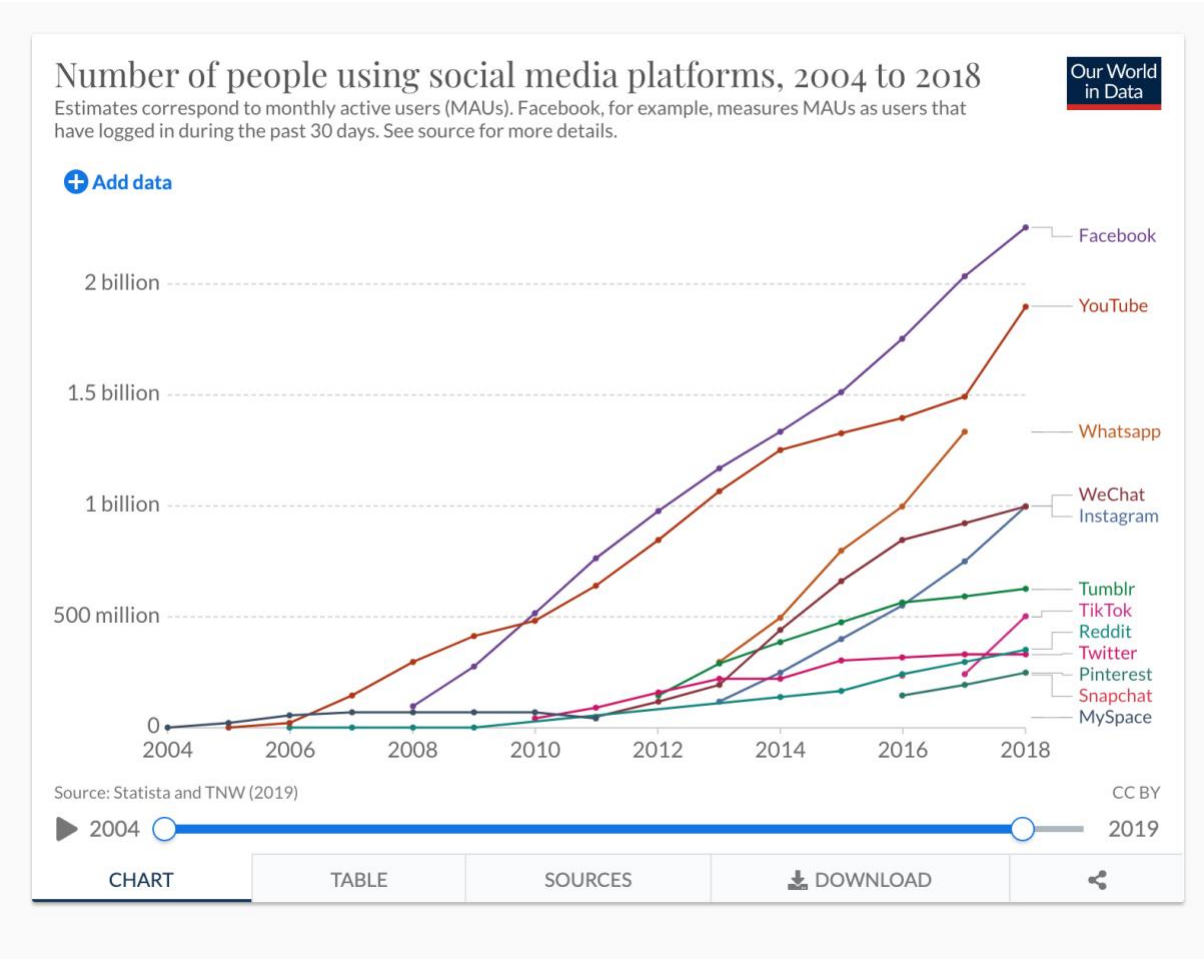


Figure 1: Number of people using social media platforms, 2004 to 2018 (Ortiz-Ospina, 2019)

Furthermore, the social media platforms are predominantly driven and used by the younger generation, as indicated in figure 2 (Ortiz-Ospina, 2019). In 2014, around 90% of the younger generation (ages 16-24) in OECD countries were engaging with social media platforms (Ortiz-Ospina, 2019).

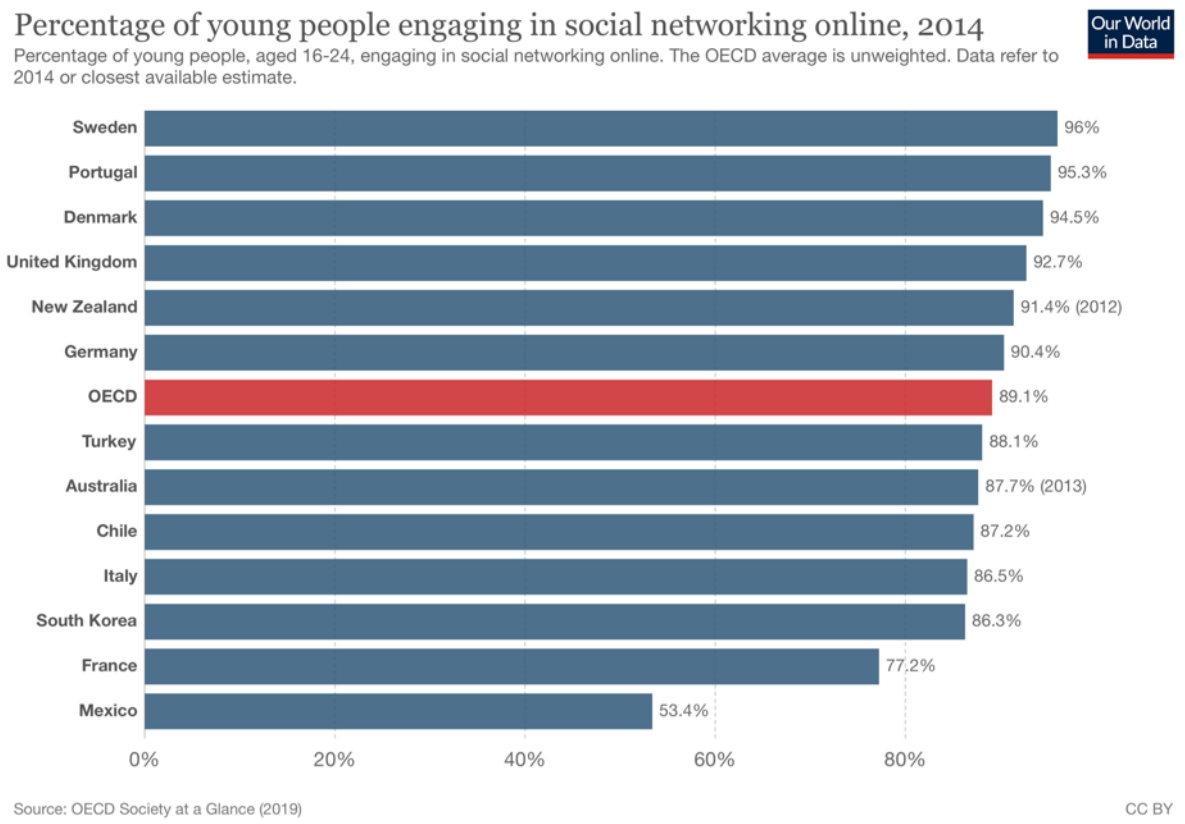


Figure 2: Percentage of young people engaging in social networking online, 2014. (Ortiz-Ospina, 2019).

With the rise of social media, adolescents and young adults have made it an integral part of their social lives. Additionally, it plays a huge role in how they might express themselves, explore and cultivate their identities (Andreassen et al., 2016). Social media thereby also gives individuals an ability and power to create any type of self, however they wish, as online platforms enable and allow this. However, is this self that individuals present online, a true representation of who they are? As explained by He & Shan (2018), self-presentations on SoMe are often warped to be more of an idealised version of who they desire to be, resulting in a more selected presentation of the self, or even a falsified version of the self. This can be through how they relay experiences on SoMe, e.g. travelling, eating good food and spending time with people. Or maybe they present themselves through sharing images, where they spend time on adjusting poses and facial expressions, to reach a desirable image. This selective self-presentation on SoMe has the ability to lead to increased social comparison, which can in turn have negative effects on self-esteem (Yue et al., 2017). Additionally, women are more vulnerable to these effects, as online culture heavily promotes and encourages self-worth that is based around appearance (Yue et al., 2017). In the

following section we will further explore which negative impacts SoMe usage can have on its users.

Never-ending content streams and the filter bubble

The observation that the smartphone and SoMe have become such a big and almost natural part of daily life is often met with a somewhat critical tone - which can also be described as techno-pessimism. Gurakan (2019) explains that techno-pessimism is alive and well. The term techno-pessimism can be explained as the negative assumptions and understandings about growing technological phenomena. In this instance, the evolution of mass-media towards SoMe platforms, as forums for global public communication. Today, techno-pessimists in particular target Social Media. For instance, it is claimed that a platform, such as Instagram, is not just a platform for mass communication, but a platform for mass dissemination, i.e. the platform is rather trying to influence or put forth an agenda by sending messages widely, rapidly and continuously. Gurakan (2019) argues that there is a distinction that must be made between the elements of SoMe that are inherent for it to function at a basic level, such as the opportunity of interconnectivity and instant global communication, and the elements that are designed to make users spend more time on the platforms. These two can be described as the essential and non-essential parts of a SoMe platform. Gurakan (2019) argues that algorithms that generate the never-ending stream of content, are some of the elements that are designed into SoMe platforms with the sole purpose of engaging the user for longer periods of time. In addition, users are exposed to what Eli Pariser has coined as our filter bubble, as he explains: "*your filter bubble is your own personal, unique universe of information that you live in online. And what's in your filter bubble depends on who you are, and it depends on what you do. But the thing is that you don't decide what gets in. And more importantly, you don't actually see what gets edited out*." (Pariser, 2011, 4.06). The longer they linger, the more they consume, the more they will be exposed to and the higher the chances are of users actually buying something that is targeted to their them through third-party advertising. This concern with use of personal data for marketing purposes and the dangers in mass political radicalisation through the filter bubble, is at the core of the critique that SoMe is receiving.

Gurakan (2019) calls for critique of SoMe, as it is both powerful and potentially the best technology at hand. It has the ability to connect the entire human population, share stories and create understanding between opposites. However, if the design of SoMe platforms is not constructively critiqued, we might find that its architecture, algorithms and interfaces may be extremely harmful to users - this might be unintentional consequences of design made with the best intentions. For instance, design that is made in a way that allows the user to stay on top of what seems relevant to them through notifications. Some users experience the application and their notifications as helpful reminders, where others might experience them as a major stress-factor.

Likes, exposure and self-worth

Another designed feature of SoMe, is the ability to 'like' and to receive 'likes'. This is a function that allows a user to show others that they appreciate, adore, like, approve, and agree with another post. In turn, it is also a way for the 'poster' to assess whether the post they created was successful or not.

A user of SoMe is constantly available, always putting themselves 'out there' for others to see. This exposure is not an image of the whole truth. Users might have a tendency to make their lives look in a certain way, through the veil of the SoMe platform. The ability to edit and direct the story of the user's own life, can create a sensation that they need to become worthy of the attention of other users. They therefore must portray a life that is equal to or better than others, in order to be judged equally. The user might make themselves an object in which others can judge as a passing thing, rather than a person equal to 'the Other'. As Sartre (1992) puts it: "*By the mere appearance of the Other, I am put in the position of passing judgment on myself as an object, for it is as an object that I appear to the Other.*" (Sartre, 1992, p. 302). On social media the user can create a self-presentation, as Yue et al. (2017) describes as being the attempts to control the narrative of their own life, by enforcing a selective presentation of their lives to real or imagined audiences.

It is often heard that there is a connection between the use of SoMe and the experience of low self-esteem. Here we argue that this might be due to some of the negative consequences of likes, bringing competitiveness and objectification of the

self through exposure of the staged self. The staged self is the version that the user will promote through their digital/online SoMe profile. Yue et al. (2017) argue that this self-disclosure may become a very negative experience due to unwanted criticism: "*As individuals engage in generally non-directed self-disclosure via social media and digital image sharing, they may consequently become the target of criticism, which in turn could affect their perceived self-worth and general life satisfaction*." (Yue et al., 2017, p. 2).

So, if the use of SoMe is so bad for the user's perceived self-worth, why don't they stop using these platforms?

Andreassen et al. (2016) suggest that this might be due to addictive behaviour that is involved in SoMe use. This is based on a large-scale investigation of the connection between SoMe addiction, narcissism and low self-esteem in Norway. Here, it is found that users of SoMe, as human beings, have natural narcissistic tendencies, meaning that they will enjoy positive feedback on their self-presentation. SoMe platforms allow these narcissistic tendencies to be nurtured at large. On these platforms it is possible for individuals to express their ambitions and show their successes to a wide audience, where they receive visible and instant rewards (ibid.). It becomes almost impossible to receive the same amount and instant feedback in other social interactions in the real world. Andreassen et al. (2016) argue that this pattern is one of the reasons why there is a rise in the addictive use of SoMe. Addictive SoMe use is measured with the Bergen Social Media Addiction Scale (BSMAS). Here SoMe addiction is defined as "*being overly concerned about social media, driven by an uncontrollable motivation to log on to or use social media, and devoting so much time and effort to social media that it impairs other important life areas*" (Andreassen & Pallesen, 2014, p. 4054).

Andreassen et al. (2016) point out that while SoMe is a great way of feeding one's ego, in order to gain better self-worth, it is not as valuable as face-to-face interactions, where individuals create long-term meaningful bonds with significant others. On SoMe, users are often strangers, who share opinions, ideas and experiences in a public forum. As Kühn et al. (2019) state: "*Social media makes teenagers in particular spend less time doing analogue free time activities and more time on their smartphone, which can be linked to increasing levels of unhappiness, loneliness, and even depression*"

(Kühn et al., 2019, p. 24). Furthermore, a consequence of SoMe use, is the lack of expected feedback or negative feedback. This, in turn, can lead to a lack of self-worth, where the user will experience that they did not live up to a standard of living that is 'likeable'.

Ubiquitous Computing and designed features of Social Media

Mark Weiser (1991) presents Ubiquitous computing as "*The most profound technologies are those that disappear. They weave themselves into the fabric of everyday life until they are indistinguishable from it*" (Weiser, 1991, p. 94). Ubiquitous computing was first presented by Mark Weiser as part of a research program at Xerox's Palo Alto Research Center (PARC). It started off as a reaction to some of the complexities around personal computing that were in the 1980-1990s, when computers were brought into the household. Suddenly, users would develop relationships with their personal computers, as it now stored personal files rather than exclusively being a functional item for work. The research led to the introduction of a new field of computer science that dealt with the cross-section between computation and the physical world. The introduction of wireless networks that could stretch globally, realised the idea of ubiquitous computing, where there were no longer limitations on where and when computing would be available. What is possible, is no longer the main challenge. The focus should rather be on the relationship between humans and technology. Weiser and Brown define this challenge as creating calmness in computing, i.e. "*If computers are everywhere they better stay out of the way, and that means designing them so that the people being shared by the computers remain serene and in control*" (Weiser & Brown, 1996).

Notifications as the enemy of 'calm'

Turkle (2017) further addresses the problem that might occur when a user of SoMe platforms can experience confusion, when the boundaries between the physical and the online realm are blurred, i.e. when we start treating the digital world as our physical reality. A general characteristic of SoMe is the ability for a user to create a persona that is solely alive in the digital world. The user can create a 'continual copresence' of their physical and online self. This might harm the users' experience and engagement with their physical world, as it becomes 'pausable'. Individuals can immerse themselves more deeply into the life they are living in the digital realm. Another

problem arises here, as the social interaction on SoMe is always ‘on’ and is ready at hand at all times. Even more so, the SoMe application might demand attention from the user through constant notifications, leaving the user with a sense of obligation to respond. This is where the term ‘Fear of missing out’ (FoMo) becomes relevant. With a rising FoMo, the users’ engagement will also rise. The designers of SoMe platforms have most likely built this with the best intentions in mind. In SoMe platforms, such as Facebook, there is a promise that the user will be connected to the happenings of the users they are friends with. A way to reach that goal is by notifying users of these happenings in real time. An unintentional consequence of these real time notifications is that they will also hinder the user in experiences that exist in the physical world. As Kühn et al. put it: “*the smartphone can become such an extension of self that when separated it provokes fear, unofficially called nomofobia, or even a lessening of self*” (Kühn et al., 2019, p. 24).

"*Information technology is more often the enemy of calm*" (Weiser & Brown, 1996). Technology is prevalent in almost all aspects of our daily routine and life, from our mobile device and personal computer, to smart devices like Alexa that realise our needs before we ourselves are aware of them. Technologies, although more helpful, require our attention at all times of the day. Our routines are often designed around the use of technologies, like our dependence on our Smartphone to navigate from one place to another, or to communicate and interact with people around us, whether for work or recreation. Our attention is constantly needed from one thing to the next; with more technologies comes more access to information. For example, with the introduction of new self-tracking technologies, we can learn more about ourselves, our routines, and a piece of technology can make this information available to us. But when do we reach a point where we experience information overload? What was once helpful has had unseen consequences to which we now constantly need to be alert and attentive to the information that is accessible around us. Can the problem of information technology be fixed by technology itself? Weiser & Brown (1996) present designing calm technologies that ‘encalm’ while informing users. The way they present this, is by focusing on the user’s periphery of attention. The periphery is explained to be “*what we are attuned to without attending to explicitly*” (Weiser & Brown, 1996). However, what is important when designing calm technologies, is that we have to use the periphery actively, and engage a user in both the centre and the periphery of their

attention. What happens in the periphery can be just as important as what is at the centre of our attention. However, being able to navigate between the information of these two, is the way in which we can inform users without overburdening them (Weiser & Brown, 1996). If a calm technology is designed correctly, by empowering the periphery, users can become more aware and familiar with their surroundings, attributing what Weiser and Brown (1996) coined as locatedness, that of feeling connected to the world around us. Further, calm technologies aren't limited to digital interfaces. Weiser and Brown (1996) present a simple example of creating locatedness in an office setting. Clues are given to create awareness of our surroundings, such as using inner windows that make people in an office aware of the going-ons of those outside of their office. For example, because of the inner windows, they can see when lights are on and off in the rest of the office, indicating whether other people are still around. This simple use of the periphery, creates context for the user and a familiarity of their environment. Their periphery is telling them things about their context, without burdening them with information. Seeing that a colleague's light has been switched off, is a subtle and calming indicator of a person having left the office, rather than receiving a message notification from that colleague.

In the infancy of ubiquitous computing, computers were focused on the excitement of interaction, but as Weiser and Brown (1996) put it, when computers are all around, we as designers must rethink the goals, context and technology of the computer, as well as all the other technology crowding our lives. The main challenge for technological designers of the 21st century, as Weiser and Brown (1996) claim, is to design calmness so that people and users of computers remain "*serene and in control*" (Brown, 1996). We see the role of interaction design, as crucial, when working with calmness, as we, as interaction designers, have the ability and need to amplify the focus on the interplay between content, context and the user - thereby consciously creating a sense of calmness. In the following, we will be looking into the interaction form of Social media based on the three form elements in the practice of interaction design (Vallgårda, 2014). We will strive to understand how calmness (or the lack thereof) in the design of SoMe plays out and affects the user of this piece of interaction design.

The three forms of interaction design

If we look at the phone as a physical manifestation of our use of SoMe, we might notice that the phone has become an almost inseparable part of our person. You might find that a person brings their phone with them everywhere; at work, on your desk or in your pocket, while waiting for the bus, in bed while sleeping, on the couch, or when with friends. You might even see someone bring it to the bathroom. This can further be described through the three form elements in the practice of interaction design: the physical form, temporal form and the interaction gestalt (Vallgårda, 2014). It is in the centre of these three form elements that interaction design occurs.

The phone's physical form might not be extremely interesting. It is a flat square, often black and neutral in its demeanour. Vallgårda (2014) describes the physical form as the three-dimensional shape of a thing, and further suggests this to include other physical elements, such as anything humans might perceive through our sensory apparatus. We would like to suggest that this too, includes what we can see. In this instance, this would be the designed components in a smartphone application. It is the contents of the phone that makes it interesting and gives value to the user. The user is able to personalize the contents of the phone by downloading specific applications that they wish to use. So, looking at the physical form, the smartphone with its contents includes a multitude of things; glass, aluminum, plastic, a camera, a microphone, wi-fi and mobile network connectivity, and the list goes on. Further, it is sprinkled with applications as building blocks for personalised use of our tiny pocket computer. In this project we are specifically looking at SoMe applications, which are often scrollable user driven content, which includes buttons for liking, sharing, commenting etc. And while hovering over the scrollable content, we take a look at what Vallgårda (2014) describes as the temporal form. This is a term that covers the patterns of state changes that a computer or device produces within the user or its environment. In other words, the way an algorithm is programmed and the intended use designed in the architecture of a platform is significant for the user's interaction with the technology. A SoMe platform, such as Instagram, Facebook or Twitter, is designed and programmed in a way that generates never ending content for the user to consume. It is the form that happens over time, that is designed through the programmed algorithms that decides, what a user should be exposed to and to what degree. If we focus on the term

temporality, we might find this to be of significant importance in the interaction with these types of scrollable never-ending contents. A user of a SoMe platform might experience themselves as *outside of* or *forgetting of* time while consuming, as they lose the sensation of how fast or slow time passes. Further, users might experience a need for content that is not time demanding, so that they can have the experience of consuming more in a shorter amount of time.

Interaction gestalt is described as the performance of movements that the user carries out in relation to the thing (Vallgårda, 2014). This can be all from touching, listening, standing, sitting, moving etc. In this case the user is physically not doing much in the interaction with a smartphone or a social media platform. Mainly, they move their fingers in a bottom-to-top direction and move their eyeballs erratically, in order to consume as much content as possible. But if we move beyond the direct interaction between the phone and the user, we might find that the user is taking the ideas and life of the platform further into their environment than initially observed. The user might consider every moment of their life in relation to their online existence:

> *Is this **worth** posting? When **should** I make my next post? Would people **like me** more if I posted something different? How can I make my life look **better**? If I just place my food like this before eating, I can take a picture people would like. I **should** probably buy some **nicer** dishes, cutlery and glasses, so I can post pictures of my food. I should probably start working out, so I can **show** people that I have a healthy lifestyle? I should live more like him/her. Should I also start doing latté art, so I can post **nicer** pictures of my morning coffee? - but then I'd have to get up even **earlier** in the morning - I would be more tired during the day - but my followers **would think** I had even more energy, because of the latte art… If it doesn't get enough **feedback** I can just delete it later. I hope they won't **judge** me too harshly. Maybe I just shouldn't **risk it**…*

This example of an inner discussion might show how a SoMe platform combined with the physical form of a smartphone can integrate itself into every little consideration a user has during a normal day.

Approaching a design space: exploring related works

The next section will be focused on exploring related works. We will be considering other projects that address different strategies in dealing with self-esteem. We will also introduce the transtheoretical model for behavioural change (Prochaska and Velicer, 1997), as we believe that this model will support in creating routines for self-appreciation. We will be introducing mindfulness as a practice, as mindfulness will be referred to as a mitigating method or approach to raising feelings of self-worth (Roquet and Sas, 2018; Brown and Ryan, 2003; Kuhn et al., 2019; Vidyarthi et al., 2012).

Changing routines for self-appreciation

We understand that changing one's own pattern of thought and routines to be more self-promoting and self-appreciating, can be difficult to overcome. This is a radical change in behaviour, and therefore we looked for inspiration in theories on behaviour change. We were inspired by the Transtheoretical Model of Health Behaviour Change (Prochaska and Velicer, 1997). We will be referring to this as the Transtheoretical Model. This model consists of six stages; precontemplation, contemplation, preparation, action, maintenance and termination. The model is created on the basis of leading *behaviour change theories* at that time. The model is most often used in behavioural changes in areas such as substance abuse, smoking, social anxiety, eating disorders and as part of other preventative practices.

The six stages in the Transtheoretical Model can be described as follows:

<u>Pre-contemplation:</u> Here, people are not intending to make any change. They might deny that there is a problem in their behaviour that could need change. Moreover, they might be overcompensating with rationalisation of their current behaviour. At this time, they might not be aware of the consequences of their behaviour.

<u>Contemplation</u>: In this stage, people are willing to make changes in the foreseeable future (but not right now). They will be weighing out the pros and cons of making a change to their behaviour. Some might become stuck in this stage.

Preparation: This is the stage where a person might start to take action towards changing their behaviour. One might see that people start looking into how they could change their behaviour and ways in which they can improve themselves. They are preparing for action by investigating their possibilities.

Action: People have made changes to their lifestyle within the last 6 months. This is where others might see that the person is actively making changes. This is why behaviour change is often linked to a specific action, even though there are several stages before this is realised.

Maintenance: This is a crucial stage in the Transtheoretical Model. Here, people are to maintain the changes they have made to their lifestyle. This can be difficult, and the chances for relapse are likely in this stage. Moreover, this stage can take between six months to several years. Relapse is a natural part of this stage, and will set a person back to one of the previous stages. Relapse does not mean that change is impossible for a person, it only stands as proof of the difficulty in behavioural change. Encouragement from others and towards one self is vital in this stage.

Termination: This stage is characterised as the complete deletion of a bad habit - as if it never existed.

For this project, we have not followed this model to the last detail. We have only drawn inspiration from three of the stages, where others have been neglected, as they were not found relevant. Our focus has been mainly on *preparation, action* and *maintenance*.

Within these three stages, there are certain elements that are of significant importance. These include *consciousness raising, self-revaluation, self-liberation, social liberation and counter-conditioning* (Prochaska and Velicer, 1997). There are other elements to these, but they are solely related to substance abuse and therefore not seen as relevant to this project.

Consciousness raising is about becoming aware of the causes, consequences and cures for particular problems. Interventions that instigate consciousness can include

education, feedback, confrontation and interpretation. *Self-reevaluation* concerns both cognitive and affective assessment of one's self-image. *Self-liberation* is about believing in ones' capability for change. Setting resolutions and multiple choices for success are some methods used in self-liberation. Further, *social liberation* is about an increase in social opportunities for upholding the changes in behaviour, i.e. no-smoking zones, healthier food choices in cafeterias etc. Finally, we were inspired by *counter-conditioning*. This concerns a person's ability to substitute unwanted behaviours with healthier behaviours. For example, relaxation counters stress, exercise counters lack of energy and being around friends counters loneliness. (Prochaska and Velicer, 1997).

Why mindfulness works

Shapiro et al. (2006) developed a model for why and how mindfulness works, through what they have coined as '*The three axioms of mindfulness'* (figure 3). The three axioms consist of: intention, attention and attitude. Shapiro et al. (2006) further emphasise how these three axioms are not separate processes, but rather interwoven aspects of a single process, where mindfulness is this said process.

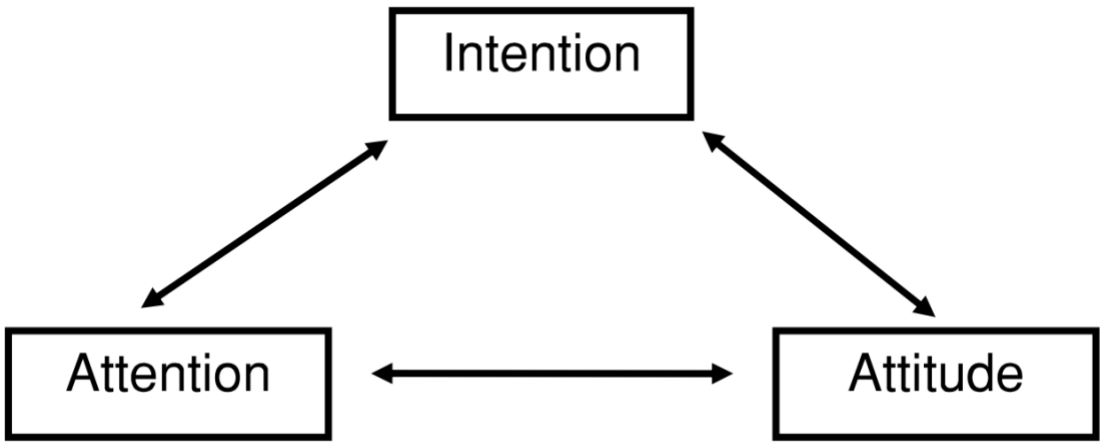


Figure 3: Three axioms of mindfulness (Shapiro et al., 2006)

The first axiom, *intention,* is essentially that a user has set a goal or personal vision for themselves, i.e. understanding why they are practicing mindfulness. Shapiro et al. (2006) further explains that the value inherent in intention, is that it is dynamic and evolving. The second axiom that is fundamental to mindfulness is *attention*. Essentially, this can be understood in the way one should bring focus to the here and

now, moment by moment. Being able to attend to and observe our internal and external behaviours, is what brings value to this axiom. The third axiom, *attitude*, is crucial to mindfulness being effective. Shapiro et al. (2006) describes this axiom as being the quality of awareness, e.g. having a critical vs. a compassionate quality towards oneself. This is what is meant by attending to the present moment without judgement or in a particular way. Especially this last axiom, is why mindfulness is effective when dealing with the theme of self-esteem. The ability to treat ourselves and our experiences with compassion, is part of what strengthens our sense of self-worth and heightens our self-esteem. The dynamics of treating our experiences with compassion, is not to disregard the negative experiences we might have, but to acknowledge and understand them, and then let them go, as Shapiro et al. (2006) explains, "*Through intentionally bringing the attitudes of patience, compassion and non-striving to the attentional practice, one develops the capacity not to continually strive for pleasant experiences, or to push aversive experiences away*" (Shapiro et al., 2006, p. 377).

Mindfulness practices have become increasingly popular, and are available in a variety of ways. Some individuals take an unstructured approach towards applying mindfulness in their everyday life by paying attention to what they are doing in the moment, through all senses - touch, sound, smell, sight and taste. Other common practices, and a reason why mindfulness is often linked to meditative practices, is focusing on breathing, which is a somewhat easier way to begin with mindfulness. In recent years, mindfulness practices have become more recognised and have therefore also led to more structured approaches within the field. Such practices include guided mindfulness through courses, workshops, internet programmes, one-to-one sessions and applications (Birtwell et al., 2019).

As interaction designers, there is a great potential design space within working with mindfulness. Especially, as it builds well upon the idea of calm computing, as explained previously. In a more technological context, calmness is sought. The origins of mindfulness can be, among other things, attributed to the search for calmness that users don't necessarily experience. The idea of wanting and seeking mindfulness, could be a result of not being able to focus on the moment when we have an assortment of things that either need our attention, or that we can pay attention to.

Challenges of mindfulness in smartphone applications

Roquet and Sas (2018) carried out a more nuanced auto-ethnographic and evaluation study of qualities within some of the most popular iPhone mindfulness apps in the UK, e.g. apps like Headspace and Calm. Mindfulness is a practice in which we try to actively focus our attention, in a particular way, on the present moment, and without judgement (Shapiro et al., 2006). Practicing mindful attention is helpful in interrupting negative thought patterns, in order to react to experiences in a more positive and helpful way. It has been proven to specifically be helpful for individuals who deal with anxiety, stress or depression, in that it makes individuals more 'in tune' with their emotional state and thereby able to alter them (Brown and Ryan, 2003). For example, when individuals experience negative thoughts, that often spiral into further negative mindset, mindfulness can help to recognise that this negativity is temporary. In the study carried out by Roquet and Sas (2018) they found that there was limited diversity across the applications they analysed. Practically all of the applications supported guided meditation. However, they found that most of the applications had audio guidance through meditation in what they define as 'extrinsic'; that of being dependent on an outside person or process. Furthermore, they found that the majority of applications required the user to meditate daily for 10 minutes, in a quiet place (preferably the same), while sitting comfortably (Roquet and Sas, 2018). Moreover, they found that applications are driven by messages (verbal and audio) to the user, that either keeps the user focused on the exercise or tries to explain the concept of mindfulness, i.e. what they are trying to do. Roquet and Sas (2018) also conclude that no applications provided tools to measure their performance during or after the mindfulness meditation. The main takeaways we see in working with mindfulness applications is, like Roquet and Sas (2018) argue, to move towards tangible interfaces, rather than limiting to a mobile interface.

Creating mindful interactions: working away from the smartphone

In 2019, Kühn et al. carried out a 2 week study, in which they explored how they could create more mindful interactions within smartphones. This came as a reaction to the negative effects of the competitive nature around the 'attention economy'; essentially, user attention and engagement on services in mobile applications have great financial value. They describe it as "*mindless consumption and meaningful fulfilment*" (Kühn et al., 2019, p. 24). For example, the attention economy is heavily taking advantage of

features such as the autoplay feature on Netflix and YouTube or never-ending scrollable content, as mentioned earlier. Kühn et al. (2019) recognise that the consequence of the attention economy have challenged a users' ability to be mindful on screen-based devices, as they explain: "*plurality of functions offered by smartphones and the variety of use-situations create complex human-computer relations*" (Kuhn et al., 2019, p. 24). They found in their study that notifications in a smartphone evoked an urge to respond for the users. Additionally, although when users were away from their smartphones, they felt 'freedom', they also recognised the feeling of 'FoMo', as we described earlier. Similar to our own research, Kühn et al. (2019) have recognised and become aware of the link between SoMe and depression/unhappiness in adolescents. However, they saw that although user's felt dependent on functionalities within their smartphone, they also felt supported by it. Therefore, their aim was not to remove or lessen the relationship between the smartphone and its user, but rather create a more meaningful and mindful experience between the user and their device. Their prototype consisted of a reconfigured user interface in which they, among other things, made it more difficult to access content, such as having to search for the specific thing you want, a prioritisation of messages being sent or having to add desired time frames for app interactions. The most important design outcomes that were found through this project, were that although the interface changes made it more difficult to interact with the smartphone, it did allow for the user to spend more time away from the device. They describe this as a "*detour and micro-boundaries to lead to constructive unfriendliness*" (Kühn et al., 2019, p. 27). This so-called detour then allows for a user to "*break routine and prevent circular compulsive use and automated impulses*" (Kühn et al., 2019, p. 27). Most importantly, they found value in focusing on creating balance between people and technology. It isn't about rejecting or embracing technology, but encouraging users to reflect on how technology impacts their everyday life.

Happy Bits: inspiration for our methodological framework

In 2018, Paay et al. carried out a 6-week cultural probe study, in which they tried to understand current practices in managing self-esteem. This was followed by a co-design project to create digital artefacts that could help with improving self-esteem. They then deployed these artefacts as technology probes for 4 weeks with 6 young adults. Their motivation for choosing to do a cultural probe study was inherent in its

value of being able to work with personal and private matters, such as feelings of low self-esteem. For their approach, they chose to work with participants who self-identified as having low self-esteem, with an even distribution of male and females. The tasks they created for the probe kit were focused on getting their users to reflect on their experiences with self-esteem, how and when they felt it and how they cope with those feelings. Examples of tasks included mood-logs and fill-out-the-blanks assignments. The probe study was accompanied by semi-structured interviews every second week. The findings gathered from their probe study were presented in a framework that they used in their workshop to inspire design ideas for their participants. The insights they gained from the design ideas were as such: self-esteem is highly individual, self-esteem issues aren't something they want to be visible in their environment, users struggled in giving compliments to themselves and the users weren't always in need of 'compliments'.

We see the main challenges in this project, as potential in future design research. Most importantly, how can we, as interaction designers, address the subjectivity and individual complexity of self-esteem? Additionally, feelings of low self-esteem are, as we have discovered through our research, habitual in nature and a natural emotion that everyone experiences. It is a consistent feeling for some and a fleeting feeling for others. Paay et al. (2018) took the approach of only addressing and placing focus on the negative emotion of self-esteem, as a way of understanding how individuals can address and manage this emotion. Finally, the probe study didn't engage with others and or their surroundings. They focused on self-esteem and managing it, as a one-person activity, rather than exploring the complex dynamics that play into why and how people feel self-esteem.

Sonic cradle: interactive design that fosters meditative experiences

A goal of mindfulness, in its essence, is to attempt to create an awareness of our 'inner world', by observing, understanding and listening to our thoughts and feelings. Various technologies have attempted to address and support individuals in being mindful. However, as Vidyarthi et al. (2012) understood, these systems were wholly ineffective. Their main motivation is that they see that most interactive technologies require the attention and continuous engagement of their users, which has proven ineffective in managing stress (Vidyarthi et al., 2012). They saw that mindfulness meditation had

been a proven therapeutic tool for psychological self-regulation, and interaction designers were beginning to incorporate mindfulness principles into stress management applications. As a result, they designed the Sonic Cradle, "*a chamber of complete darkness where users shape a peaceful soundscape using only their respiration*" (Vidyarthi et al., 2012, p. 1). The goal was to assist users in reaching a heightened awareness of their own thoughts, to help manage stress. Their main challenges when designing the Sonic Cradle was in creating a sense of immersion: "*generating complete engagement without simultaneously providing new sources of distraction*" (Vidyarthi et al., 2012, p. 3). Sonic Cradle mimics the experience of immersion by suspending their users in a dark chamber, i.e. with no visual distractions, where users can control a 'healing soundscape' through their breathing. The Sonic Cradle embraced mindfulness in their search for immersion, however they did see that the experience was less engaging over time, especially when individuals felt that they were losing control over their immersion. This, however, would match up with qualities of meditative experiences in that their users would over time lose track of their 'reality' in that they became disengaged with the physical reality and were just focusing on their breath and the noise in the chamber.

Defining our design space

In this next section we will try to explain how and where we will place our design efforts, in order to make a meaningful design contribution. We will attempt to address the aforementioned related works and what we have found within their research that gives our project value and can help address our research question. This will in turn, form the design space that we wish to work within and how we potentially can contribute to this design space through our workbook.

Designing with mindfulness practices

Mindfulness Based Applications (MBAs), as described by Roquet and Sas (2018), although they help some, still create a dependence on an application. Since mindfulness is essentially an intrinsic experience, i.e. that of being independent of outside elements, we see value in trying to move away from the dependence that might exist on a phone or screen. In order to assist with mindfulness practices, we believe there is value in working with an artefact that is beyond the traditional

application and its limiting capabilities. Kühn et al. (2019) worked with creating mindful interactions within the smartphone (beyond an application), by trying to change the interface and the way we interact with the interface. However, what we found most insightful, was the value user's experienced when they were 'away' from their smartphones. They recognise that designers should try and work with creating balance in HCI, which is why we see value in moving further away from the smartphone. We want to, as also expressed in our research question, try and motivate and assist users in practicing self-appreciation possibly beyond the use of a smartphone. We see that there is an interesting design space within interaction design, in incorporating technology in a way that feels almost non-technological.

The value inherent in the Sonic Cradle project, was in the creation of immersion. In MBAs, there seemed to be a specific way to address mindfulness and mindfulness meditation, where almost all applications were around guided meditations. The Sonic Cradle project was important in acknowledging different ways in which we can address mindfulness and explore meditative aspects in our project. Rather than looking at guided meditation through speech, there is also potential in using sound and breath, with immersive properties.

Designing for individuality

The Happy Bits project was insightful when it came to inspiring a methodological framework for our project: how will we do design research around self-esteem? We saw value in using a cultural probe study, as it allows for individuality and subjectivity in design. The Happy Bits project was focused on creating a single solution that would help with self-esteem. Our goal is to try and incorporate the diversity in how young adults experience and manage feelings around self-esteem. Furthermore, based on our research, self-esteem is something habitual and a natural emotion that everyone experiences. Therefore, we didn't want to focus this project around the negative emotion that low self-esteem can prompt. We wanted to make it natural for young adults to incorporate self-appreciation into their daily routines. While Happy Bits focuses on how to manage negative feelings of low self-esteem, we wanted to engage with the aspects where users feel good, i.e. self-appreciate, as we see that this is where good habits can be built. Furthermore, we see that Happy Bits lacks in its consideration of the environment around users and their significant others. In

addressing low self-esteem directly, as something we have to handle by ourselves, they implicitly create shame around the matter. The more we think of the matter as something we have to deal with and address by ourselves, the more we will feel it is something shameful that we shouldn't share with those around us. Furthermore, literature proves that self-esteem is not only heightened by being attentive and kind to ourselves, but also being able to seek comfort and joy in the people and things that we find significant around us. To build self-esteem, it is about finding a balance between creating positive habits in our 'inner world' as well as our 'outside environment'. This again plays into the complexity of self-esteem and how we all need and crave something different.

Methodological framework

In this chapter we will explain how we are approaching our design space and problem area through a methodological framework within design research. We will be looking into Research through Design (RtD) as our framework for our design research.

Design Research

Within the field of design research, we are working specifically with interaction design. Interaction design is a holistic design discipline that encompasses a designed artifact, it's users and the context in which it takes place (Fallman, 2008). Umeå Institute of Design presented a model of interaction design research with the purpose of guiding future interaction design research, when interfacing with industry, academia and society. Fallman's Design Triangle (2008), works as a tool for better understanding the relationships and roles in the increasing multidisciplinarity of interaction design research. The simple form of Fallman's model, allows for design research activities to be placed between three extremes: *design practice, design studies* and *design exploration*. Fallman (2008) argues that what distinguishes interaction design research from other disciplines is its ability to consider all three perspectives when doing design research.

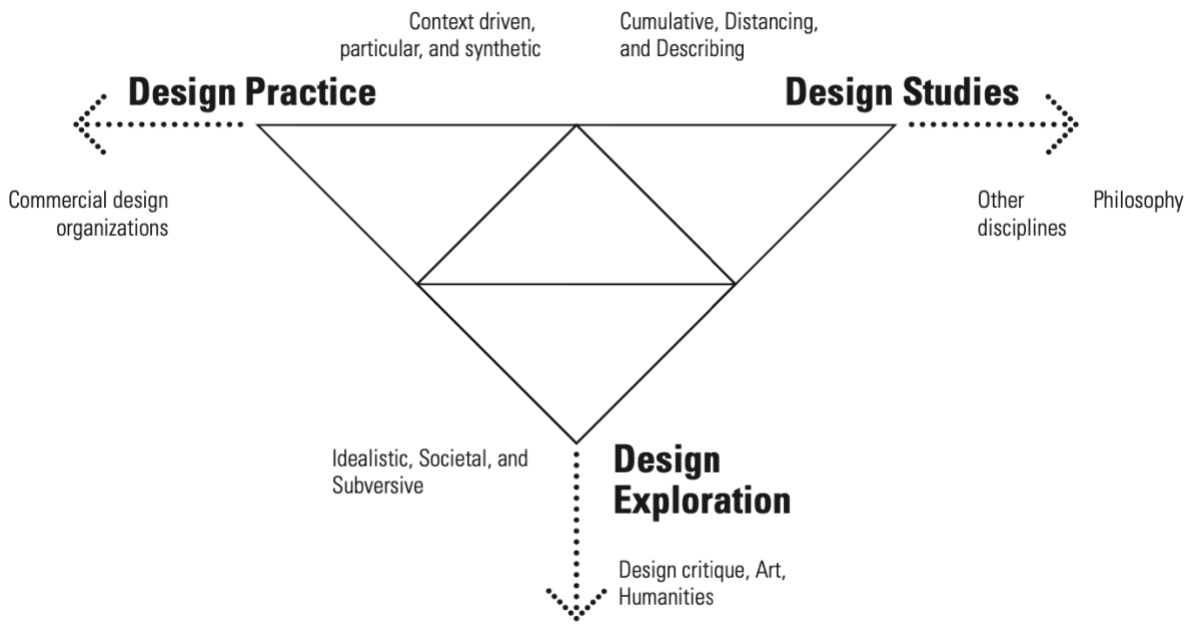


Figure 4: The interaction design research triangle (Fallman, 2008)

Design Practice represents the area in which interaction design research meets the practicality of industry. Fallman (2008) argues that in this field, a set research question is essential. This is because there are stakeholders across various disciplines within this field who have expectations and requirements for specific design outcomes, for example in the creation of a product or service. Design Studies, however, makes space for reflection and philosophical questions and discussions in order to contribute to or gain knowledge, typically within design theory and methodology. The last extreme, Design Exploration, is similar in nature to Design Practice, as it also seeks to create an artifact. However, it differs in that it is less judgemental and seek to test ideas, ask questions, provoke the existing and present alternatives to the traditional.

The value that placing design activities in the model can bring about, is providing perspective. Therefore, as Fallman (2008) argues, moving between areas, i.e. changing perspectives and acknowledging tensions between them, is where most value is achieved.

In our project, we are primarily working within the two activity areas of design studies and design practice. We place our project within design studies, as we have a desire to understand what prompts feelings of self-esteem, with an ideal of creating a solution that can mitigate feelings of low self-esteem (design practice). We are interested in better understanding the phenomenon of having low self-esteem, in order to explain why the feelings occur and how to predict and prevent them. Aspects of design practice are especially relevant, in that we aim to prevent feelings of low self-esteem, ideally, through a product or service. Especially in this case, we can identify dimensions in trying to create and change, while explaining and understanding. There are hints of design exploration in our thought process, in that we are criticising the current state of affairs in relation to SoMe usage and our compulsion to use our smartphones.

Within these design areas, focusing mainly on design practice and design studies, we are taking a Research through Design approach. Design research is commonly characterised in three ways; research about design, research for design and research through design (Zimmerman et al., 2010, p. 313). Research about Design is

understanding design as a human activity. Research for Design is about improving the design practice, often through design frameworks, design methods and design recommendations. Finally, Research through Design (RtD) is an iterative process where artifacts are made in creative ways in order to investigate potential futures.

Research through design

In this project, we are using the research approach of Research through Design (RtD) to explore what affects low self-esteem in young women, and what they do when they have feelings of low self-esteem. RtD is a research approach within the field of Human Computer Interaction (HCI) that uses design practice as a legitimate method of inquiry, whether it be around the making of a product, service, environment or implicitly gained knowledge through an artifact (Zimmerman et al., 2010).

In this context we are taking a user-centred approach, where we attempt to make our process as participatory as possible for our users. Participatory design is essentially concerned with how a design process involves the participation of the people affected by the technology that is designed (Robertson and Simonsen, 2013). It originally stemmed as a reaction to civil rights movements in the 60s and 70s, where people in Western societies demanded more say when it came to decision making concerning aspects of their own lives. This manifested into a design approach that is driven by the idea of mutual learning, where users of a design can provide unique insight and knowledge to a designer and vice versa. It is further led by principles of democracy and empowerment and that all participants have valuable knowledge. And the real value in this design approach is acknowledging that having a different voice with a different outlook, than that of the designer, is needed and important in the ongoing evaluation in a design process. The participatory design approach stands in stark contrast to the ‘rational’ and problem-solving approach to design, as it actually recognises the complexity of design problems or design situations. The concept of collective reflection-in-action, as originally presented by Schön (1983; in Robertson and Simonsen, 2013) for the individual designer, is a way to acknowledge these complexities and use it to the benefit of a design process. Collective reflection-in-action is essentially that there are continuous and collective reflections and discussions between users and designers, and that these reflections from each party are considered inherently important and legitimate and are acknowledged as part of the design process. It has the purpose to provide better understanding on how future

technologies can be embedded into our everyday environments, by letting those who navigate the environment have a say on said design. (Robertson and Simonsen, 2013)

We find RtD especially appropriate for our project, as RtD is research about the future. Here, we can research possible future scenarios through methods that help enact such a future. This was appropriate for this project because we wanted our users to engage with new possible ways of approaching their routines around self-appreciation. RtD also makes it possible to design new artifacts, where the thing itself can be seen as a contribution to theory on the subject (Zimmerman, 2010). Here, we are creating design arguments which are to be understood as a theoretical design contribution to the specific context of this project. We also find that seeing the final design outcome, as a theoretical contribution rather than a commercial product, allows us to look into solutions that are not solely made with a commercial point of view, where the artifact is designed to be sellable. We are thereby able to look into designs that can be very specific to the problem at hand.

RtD and wicked problems

RtD is often an approach that is applied when confronted with Wicked Problems. The term 'Wicked Problem', first coined by Horst Rittel in 1972, is in its essence, a problem that seems unsolvable in its complexity, as they are often ill-defined with varying explanations, with no immediate solution and are essentially unique (Buchanan, 1992).

RtD is ideal for handling these types of problems in its "*iterative approach to reframing the problematic situation and the preferred state as the desired outcome of the research*" (Zimmerman et al., 2010, p. 310). Design studies, as mentioned previously, is in the pursuit of knowledge, which can be compared to 'Research for design', which focuses on improving design practices in creating design methods, guidelines and implications (Zimmerman, 2010). In our project, we hope to bridge the gap between the creation of knowledge and designing solutions to wicked problems by using RtD. This allows us to actively use design methods to inquire about issues on self-esteem, reflect on the knowledge we gain through these practices, while allowing a design space in which we can create a digital artifact that in itself, implicitly contributes to design theory.

To further understand what a Wicked Problem is, Rittel (1972) defined ten characteristics of a wicked problem. In relevance to our project, some of the characteristics are worth mentioning. As we have stated earlier, there are many complexities involved regarding young women who experience low self-esteem. In literature, we have identified some of the long-term consequences of experiencing low self-esteem; such as suicidal ideation, depression and anxiety (Wilbum & Smith, 2005; Emler, 2001). Further, the reasons for long-term low self-esteem are abundant and very subjective to each individual. A characteristic of a wicked problem, as defined by Horst Rittel (1972; in Buchanan, 1992), is that a wicked problem is impossible to definitively define. Here, experiencing long term low self-esteem is something that most people can recognise, but looking into the casualties for low self-esteem you find that each individual has these feelings for very different reasons.
The subjectivity of what can cause low self-esteem for each individual also calls for very subjective solutions for these experiences. Another characteristic of a wicked problem is that they don't have a stopping rule, meaning that there is no a way to know whether a solution is final (Rittel, 1972; in Buchanan, 1992). This led us to work with design arguments and a workbook, which allows us to investigate several aspects of possible solutions.

As described earlier, there is a new element that can affect self-esteem for adolescents, who are especially vulnerable to comparing themselves to others; namely SoMe and the smartphone, which allows for constant self-evaluation through other individuals online. We are not stating that SoMe is the sole contributor to long term low self-esteem, however we argue that this fairly new technological element creates a layer of complexity to a problem that already has an abundance of causes. Here, another characteristic is that every wicked problem is a symptom of another higher level problem (Rittel, 1972; in Buchanan, 1992). We see problems stemming from social media usage feed into the growing levels of low self-esteem.

What influences a person's feeling and experience of self, is entirely individual. Therefore, we are, in this context, "*fundamentally concerned with the particular, and there is no science for the particular*" (Buchanan, 1992, p. 17). In order to understand the dynamics of the particularity of low self-esteem, we decided to use an explorative

approach of cultural probes as a way to break down the wickedness of the problem concerning what affects self-esteem and what works for the individual in mitigating feelings of low self-esteem.

Using Cultural Probes to understand particularity

Probes are essentially a collection of tasks that have the goal of eliciting 'clues' about people's lives and their thoughts (Gaver et al., 2004). The probes are often formed into playful kits or packages that are sent out to users over an extended period of time, where the users can explore, reflect and give their subjective interpretation on the tasks in the kit. In subsequent years, cultural probes have developed into a recognised method for inspiring design ideas and conducting design research in more personal contexts. Cultural probes have taken many different shapes and forms, for example providing disposable cameras, media diaries and maps to understand user behaviour. We have applied cultural probes in our project, in order to gain insights into our user's specific contexts. Further, cultural probes allow us to explore the long-term dynamics of working with routines. Moreover, this method allowed us to gather insights that were happening in-situ, rather than having the users explain their thoughts and actions in retrospect. Finally, the design and materiality of our probe kit allowed us to evaluate possible design features. Cultural probes were initially introduced by Gaver et al. (1999) in a project that attempted to provoke responses from elderly people in diverse communities. Their motivation for using probes was so they could gain an understanding of local communities, without creating designs that seemed "*irrelevant or arrogant*" (Gaver et al., 1999, p.22). As they explain further, "*We didn't want the groups to constrain our designs unduly by focusing on needs or desires they already understood. We wanted to lead a discussion with the groups toward unexpected ideas, but we didn't want to dominate it."* (Gaver et al., 1999, p. 22). Gaver et al.'s (1999) cultural probe study is especially interesting in that, as they explain, they weren't trying to determine the specific needs of their users, but rather gain an understanding of their beliefs, desires, aesthetic preferences and cultural concerns (Gaver et al., 1999).

Auto-biographical diaries as part of a cultural probe

As explained previously, cultural probes are structured around a users' ability to self-document, whether it be through media, images or words. To supplement a cultural

probe, auto-biographical diaries are relevant. The use of auto-biographical diaries is a process of self-documentation, in which a user reflects on actions through written language. It is a valuable tool for users to understand their own practices, while simultaneously providing others/designers with insights to their specific experiences with a product or a service (Breakwell, 2006). In the context of handling self-esteem, the method of putting words into thoughts, can not only provide an outlet for those thoughts, but allow them to pay attention to their current thought patterns. Further, we see this as an appropriate method for this project in that it can enable us to get in-depth and personal insight into our users' reflections and thoughts around self-esteem.

Evaluation

When approaching solutions for wicked problems, we have learned that it can be more valuable to explore a solution area than a single solution. As we have characterised our problem as wicked, we also find it necessary to evaluate several possible tools for mitigating self-esteem. Through our cultural probe, we attempted to evaluate possible design features, expressed through tasks and activities within the probe kit. Here, we did not find it valuable to apply traditional evaluation methods, like usability testing, as they focus on testing qualities of a specific solution.

Within the field of HCI, evaluation has played an important part. It is often focused on the user interface. This interface determines the user's likelihood of benefiting from the service system. Here, an evaluation helps identify areas where the chances of the user experiencing maximum benefits are low. The most common type of evaluation method is usability testing - these are often quite formal and structured tests of a user interface using a system or prototype. Often more quantifiable measures are used to identify whether a user interface has good or bad usability. These are measures such as completion time, amount of click (or the like), amount of help needed to complete the task. The long-term value of a digital artefact is hard to assess in this type of usability testing, which further motivates our decision in working away from traditional usability testing. A more contextual evaluation demands a longer test-period and access to the environment in which the artefact should be used. This is where we have seen an opportunity to engage with more qualitative methods grounded in

ethnographic techniques such as cultural probes and auto-biographical diaries. (Löwgren & Stolterman, 2004)

In this project we are thinking of evaluation as a long-term contextual activity. Contextual inquiry works to assess results of a design process. In this line of thought, a product is never really finished. Development and evaluation must be done iteratively, even after deployment of a product or service. Here, the line between the use and design of a product is not clear, as both happen simultaneously. This is what Henderson and Kyng (1991; in Löwgren & Stolterman, 2004) has coined as *continuing-design-in-use*. The cycle of continuing-design-in-use starts when the user begins making changes to an artefact, in order for it to better fit into a context. These changes must be documented along with argumentation for the change. The next step is that users learn about the change and incorporate it into daily life. Here, it is important to provide feedback to the initiator of change. Finally, the role of the evaluator is to support and facilitate these local processes of change, rather than being an expert who initiates the change from an outside-in point-of-view. (Löwgren & Stolterman, 2004, p. 93) In our project, evaluation as a long-term contextual activity is done through the use of our cultural probe which was deployed for four weeks. As a final evaluation of the specific features in the cultural probe, we carried out a participatory workshop in which user's experiences with the probe kit worked as evaluations.

Evaluation through a participatory workshops

A major strength of a participatory workshop is the ability to gain insight into user experiences and needs. Here, users work together to create detailed accounts of their desires for future practices through various tools and techniques (Robertson and Simonsen, 2013). In our project we decided to frame the final evaluation of our cultural probe as a participatory workshop, where the users could collaboratively evaluate their experiences with the probe kit. In order to instigate this evaluation, we prepared several techniques that would help carry the users' evaluation through an online workshop. These techniques were classic brainstorming, dot voting, and prioritisation of 'chance for success' of specific design features as it allows for reflection-in-action. Participatory workshops are often applied to "*propose, represent, interrogate and reflect on different aspects of the developing design continually throughout the*

process" (Robertson and Simonsen, 2013, p. 9). Design activities or design games are valuable to participatory design, as they help "*visualise, simulate and experiment with selected elements of envisioned technologies, in use, prior to their development and implementation*" (Robertson and Simonsen, 2013, p. 9). Here, we allowed the users to bring forth which elements of the cultural probe they deemed valuable to develop further. We encouraged them to focus on this through brainstorming and prioritisation of issues, based on their experiences with the cultural probe elements.

Analysing our empirical data

In this project we have worked with multiple datasets; data from our cultural probe (consisting of journal entries/auto-biographical diaries, images and notes from unstructured interviews), data from a participatory workshop and data from a questionnaire. In order to analyse this data, we decided to focus our attention on the empirical data from the probe kit and the workshop, as our questionnaire had a slightly different focus, and was not part of our participatory process or part of evaluating design features. In order to approach the body of empirical data gathered in this project, we decided to make use of coding in order to find categories and themes that would later assist us in the creation of our design arguments.

Coding our empirical data

Coding is a method stemming from grounded theory. Grounded theory is regarded as one of the first methodological approaches to qualitative research. It offers the application of codes to data through an iterative process which should lead to the generation of a theory (Saldana, 2009). An iterative process should be applied when coding, as the first attempt is rarely an exhaustive representation of the dataset. The researcher might see new patterns as they will take several coding iterations on the same data. In our project this theoretical outcome is the set of design arguments. These will be applied in a design workbook.

A 'code' is a word or short phrase that captures the essence of a portion of language-based or visual data (Saldana, 2009, p. 3). The data in our project is based on journal entries, workshop notes, questionnaires and visual materials (i.e. user generated

images from the cultural probe). Coding is not an exact science; it's primarily an interpretive act. First impression phrases can be codes. These are often derived from an open-ended process called *initial coding*. In our project, we have used initial coding as part of becoming familiar with our data. Some of the codes used here, were based on specifying which design elements a data entry would touch upon. A later iteration on this was to make codes based on the emotional experiences the users had with these design elements. When we reflect on a part of the data and try to understand its meaning, we are *decoding*; when we finally decide what its label should be in order to appropriately describe its essence we are *encoding*. When coding, one might consider varying patterns for identifying which codes to apply.

Our codes are based on our approach to our research problem. Especially our initial research, which is mainly represented in our problem analysis, has been the stepping stones for creation of our codes. As Saldana (2009) describes, the researcher must wear an analytical lens when coding. This analytical lens is covered by a certain filter, based on the researcher's background. Here, it is argued that no research is truly objective, "o*ur analysis and interpretation – our study's findings – will reflect the constructs, concepts, language, models, and theories that structured the study in the first place"* (Merriam, 1998, p. 48; in Saldana, 2009, p. 7).

Analysis can be understood as the search for patterns and later for ideas that might explain why patterns occur (Saldana, 2009). This, for instance, is what happens when codes become overarching categories. For instance you'll find that several codes might belong to an overall category in our analysis. In our analysis you will find that the categories created in this project are what formed the foundation of the design arguments. Finally the outcome of our coding has ultimately been through the creation of a design workbook.

Design Workbooks

We are using design workbooks to synthesise our findings and research into a communicative body of work. This will help in making complex and abstract design arguments more concrete. As Gaver (2011) explains, "*a given idea does not have to bear the full weight of a project's expectations, but there is still an obligation to act, and a requirement to concretise ideas, which are crucial in moving beyond vague*

notions of possibility" (Gaver, 2011, p. 1553). Design workbooks are essentially a collection of design proposals drawn together to investigate options for design (Gaver, 2011). Moreover, they can be classified as a design method, rather than being sketches of final ideas, as they can be used iteratively in a design process to suggest approaches and options in designing for a given situation that are then subject to change. In our design workbook, we have presented our design proposals as suggestions for ways in which design arguments can be applied. Gaver (2011) explains that a design workbook can serve a purpose that is beyond a tool for reflection; the workbook in itself, can become a subject of publication in its own right. Design workbooks should be treated as a collection of ideas rather than a representation of final designs (Gaver, 2011). Our design workbook will thereby serve as a collection of prototypes, where each proposal serves specific design arguments. These prototypes have been among other things, constructed through wireframing, mock-ups and scenarios.

Process

In order to describe our process we might look at a process model such as design thinking, which consists of five stages: *empathise, define, ideate, prototype* and *test* (figure 5).

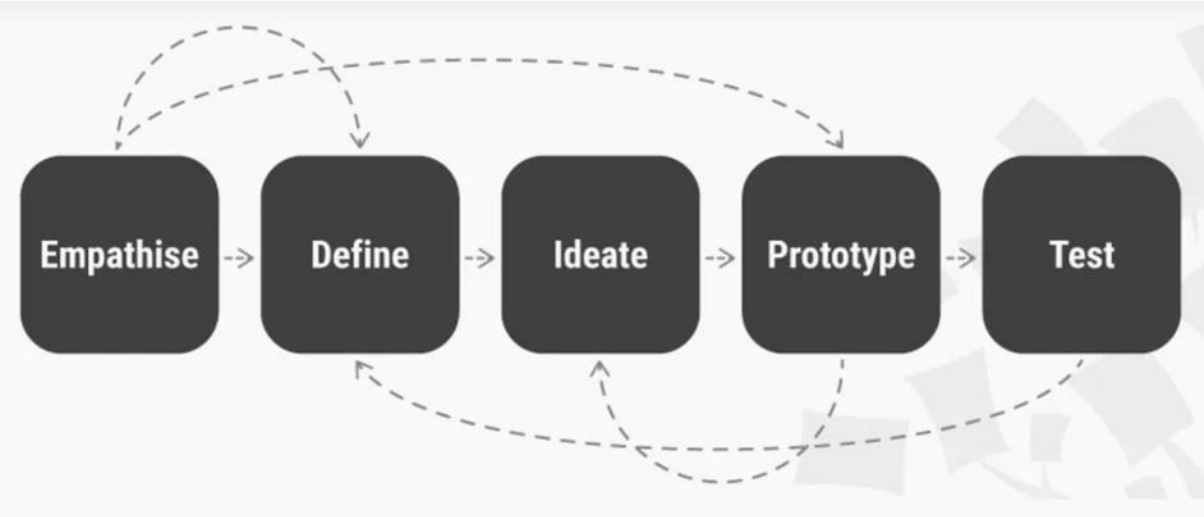


Figure 5: Design thinking model (Interaction Design Foundation (IxDF), n.d.)

In our design process, it can be argued that we have made heavy use of design thinking. This is represented in the fact that we have applied the design thinking stages iteratively throughout the entire design process. Empathise and define are focused on setting the stage, where we attempt to understand, dissect and define the chosen problem. The next two stages, ideate and prototype, are rather focused on creatively thinking of and manifesting solutions to the determined problem. The last stage is the test stage, in which designers evaluate the created solutions. In the following, we will elaborate on how this process has played out, and how we have applied the aforementioned stages of design thinking throughout.

We first carried out an initial desk research, which is represented in the first chapter of this report. Thereafter, we created a questionnaire in order to validate our newly found knowledge on the subject, applied in a local danish context. After receiving the answers from our questionnaire, we set out to design and plan our cultural probe, which will be presented as a pictorial in this chapter. After the probe had been deployed with our users for four weeks, we held a participatory workshop with the same users. This was done in order to evaluate the design elements and the users'

experiences with the probe. Hereafter, we set out to code and analyse our empirical data, which finally led us to creating prototypes and sketches that would end up in our design workbook.

Initiating the design process

Our design process began with recognising a problem that was based on personal experience, related works and general media coverage on rising levels of loneliness and low self-esteem in danish youth. To investigate this problem, we initiated our research process with desk research where we focused on questions such as: What is low self-esteem? What are the consequences of low self-esteem? Has the amount of people experiencing low self-esteem really been rising? Is the problem wider than the danish context? Who most often experiences low self-esteem? Are there correlations between low self-esteem and social media use, as stated in the media coverage on this topic? These initial questions were part of the first iteration of the empathise-stage in the design thinking process, by seeking a deeper academic understanding of our problem.

These questions became the basis of our investigation. Further research on the topic led to a structured literature review, as presented in the first chapter of this report. Through our research, we became aware of projects that had a particular focus in designing digital artifacts that strengthen self-esteem in young women. We sought inspiration within the methods of these projects, and took their advice for future work into consideration, hoping to contribute to a larger body of research in this area.

After our desk research into low self-esteem, social media usage, consequences thereof and related works, we framed our research question. This is where we started engaging with the define-stage of the design thinking process.

Validating in a local context through a questionnaire

Beyond our initial desk research, we needed further understanding of the problem, but this time how it manifests in our immediate network; specifically, to understand what self-esteem means in a danish context. This was a further iteration on the empathise-

stage. Moreover, we wanted to see if we could see a link between low self-esteem and social media use in our specific context. Finally, we surveyed which methods people use to mitigate feelings of low self-esteem, in order to incorporate and be inspired by these methods for the formation of our cultural probe.

The questionnaire (appendix A3) was answered by 128 persons. Of these, 62,5% were women, the majority of the surveyed people were in the age group 19-26 by 70%. The first part of the questionnaire was a demographic inquiry, in order to see trends across gender, age, occupation and relationship status. We then proceeded to ask our respondents how they perceive self-esteem in their own lives, what prompts those feelings and what mitigating techniques they use if/when they have low self-esteem. Moreover, we also asked questions pertaining to mindfulness, to gauge the awareness around the technique. Further, we asked how and if they seek help from significant others on matters of self-esteem. The final segment of the questionnaire was focused on their experiences with Social Media.

Who are our users?

As mentioned earlier, negative stress in adolescents is often linked to life changes. Our social confidence and school abilities are exceedingly challenged in an academic and social context, such as in high school and university. Our self-regard is especially challenged when we, as individuals, have to figure out what we are good at, what we want to spend our life doing and what will give us value. We want to focus on individuals who are in a transitioning period of their life, where they are trying to figure out their identity, while navigating social and academic pressures. This is especially true to individuals who have left high school, and no longer have a set plan for what their life will look like. We have therefore chosen to work with individuals in the age group of 19-25, who have left high school, and are thereby independent, whether they manifest their independence in taking gap years, working or studying.

We have chosen to delimit our users to women. In a study carried out by Stein et al. (1992), it was found that there are differences in how males and females gain self-esteem. They found that a female's self-esteem is primarily gained in a social dimension, i.e. through relationships, while males gain their self-esteem in a

performance dimension, i.e. through feelings of success (Encyclopedia n.d.). In its essence, "*it can be seen that boys gain self-esteem from getting ahead whereas girls gain self-esteem from getting along*" (Encyclopedia, n.d.). We find this problematic, as by this idea, women aren't encouraged to make strong claims about their self-worth, as Emler et al. put it: "*Boys are encouraged to be assertive and self-promoting, girls are not*" (Emler 2001). By this notion, females are more likely and prone to struggle with their self-worth, as they aren't taught to embrace feelings of high self-esteem. Further, desk research has also shown that women are more prone to showing signs of addictive use of SoMe (Andreassen et al., 2016). Especially in this, we see that there is a greater design opportunity to work with women, in that we see it as being of a greater need with women, as they might find the outcome more valuable.

Our assumption is that feelings of low self-esteem and low self-worth is a phenomenon that is common to everyone, across gender and age. We are therefore not excluding individuals based on whether they have a diagnosis. In our project, our aim was to create and help anyone with feelings of low self-esteem, where some people might benefit more than others. We also wanted to ensure that we weren't touching upon mental illnesses with young individuals, as we are neither experts nor have we contacted experts in the course of this project. In our project, we regard low self-esteem as feelings of low self-esteem rather than low self-esteem as a whole. This is because we recognise that all can experience feelings of self-esteem, but not consider themselves as having low self-esteem.

Recruiting participants

As part of recruiting participants to be part of our empirical work in this project, we have made use of the snowballing effect. Here, we looked into our own immediate network, and identified three young women, who fit our user description. Our accept criteria for users were as follows:

- Women
- Aged 19-25
- Located in Denmark
- Experience feelings of low self-esteem
- Are interested in improving their self-esteem

- Have undergone a life transition in recency, i.e.:
    - Started university/education
    - Moved away from home
    - Taking a sabbatical after high school
    - Experienced a break-up (hardship concerning significant others)
    - Other life transitions, that can impact one's self-concept (personal perceived identity)
- Had to be able to participate in a four week 'probe kit investigation'

Initially, we found two users who fit our criteria. These two led us to four more users, out of which one was not able to participate in a four-week project.

Finally, we had five participants for a probe kit investigation. The users were briefed on the project and our expectations for their participation. In this brief, we were careful not to tell them what we wanted them to do, but rather explained to them that they were allowed to use the kit in a way they felt best fit into their daily practices. All of our participants have been anonymized and given a new name for use in this report. They are identified as follows:

- Mette, 24 years, student, lives in Århus (DK)
- Nanna, 19, student, lives in Århus (DK)
- Camilla, 23, student, lives in Lyngby (DK)
- Simone, 24 years, student, lives in Århus (DK)
- Eva, 24, student, lives in Århus (DK)

Cultural probe: the 'fuck-low-self-esteem-I-am-awesome' kit

After our initial desk research we looked into the creation of a cultural probe. We found this method useful in investigating this problem area, as we had uncovered that low self-esteem is a natural and common emotion that can occur with anyone, at any given time. In addition, we were inspired by Paay et al.'s (2018) Happy Bits and the way they used cultural probes in their project. We saw potential in gathering somewhat 'live' data from our users, in that we would allow them to document their experiences and reflections on certain emotions in the moment they were occurring, rather than having

to recall them in retrospect. We also knew that building better self-esteem was something that demanded time. Building self-esteem is about building new routines and experiencing a sense of change in one's well-being. With a cultural probe, we were able to let the investigation run over a longer period (4 weeks). We do believe that a longer time period would have given better results, but 4 weeks was the longest we could spare in the scope of this project.

For the creation of our cultural probe, we were inspired by the Transtheoretical Model (Prochaska and Velicer, 1997) and its stages. We will mostly be addressing and making use of the stages of preparation, action and maintenance. With these stages in mind we articulated the following goals for the cultural probe:

1. **We want to help users identify what works best for them in terms of boosting self-esteem and allow for edits and customization.** This is linked to both preparation and action. We believe it will be easier for users to maintain new behaviours if there is a personal motivation for these actions. For our first goal, it was very important to create the space for users to investigate their own preferences in terms of doing self-care. For some, promoting self-esteem through self-care is a foreign thing to do, and many have tried the one-size-fits-all methods through smartphone applications, such as Calm or Headspace, which we addressed earlier. We believe that self-care happens through differing activities for various individuals. As for promoting self-esteem, one must understand which parts of one's global self-esteem is important and will have the biggest impact (Rosenberg, 1995).
2. **Create a space for mindfulness and reflection for each element of the cultural probe.** In order for our users to raise their consciousness, we need to help them reflect on their new behaviours versus old behaviours. Further, we have seen in related works that mindfulness has been a frequently used method for reflection and appreciation. We see appreciation as a core contribution in the maintenance of new behaviours.
3. **The cultural probe should assist the users in building routines around self-care**. This goal has a clear link to the maintenance of behavioural change.

The creation of the probe kit initiated the stages of ideate and prototype in this project. The ideation stage was inspired by related works and feedback from our questionnaire. We found several techniques for meditation and behaviour change that we wanted to explore further through the cultural probe. These will be elaborated in

the pictorial. In order to create the material aspects of the cultural probe, we made use of materials that were accessible and very malleable, such as paper and cardboard (Images 1-3). In order to test different interaction forms we also included a stethoscope, an egg-timer and a cookie-jar. All these artifacts are common and familiar to users - but in the probe kit we redefine their function to address self-appreciation.

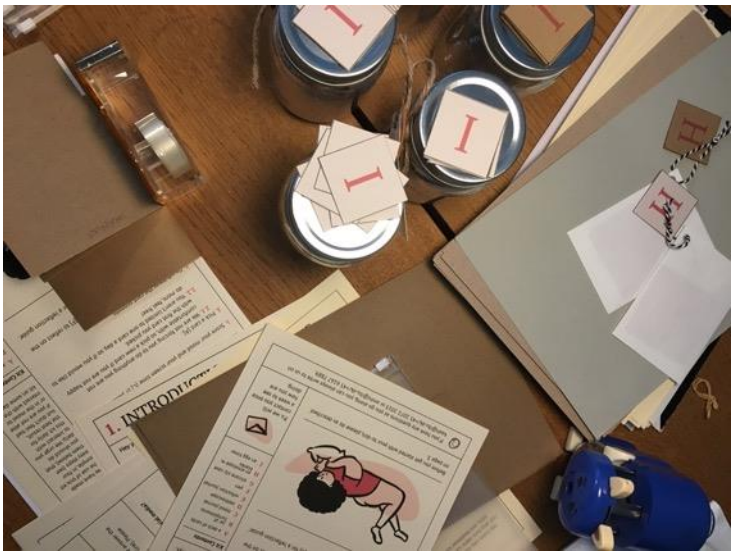


Image 1: Images of our workstation during development of our cultural probe (I)

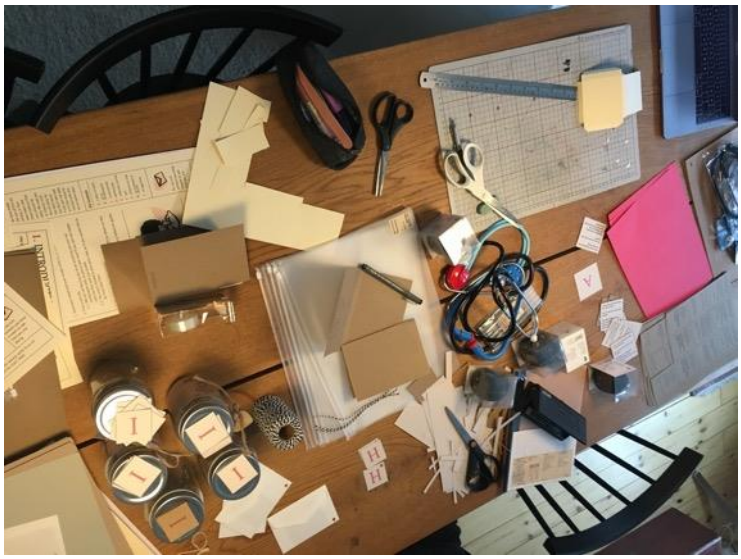


Image 2: Images of our workstation during development of our cultural probe (II)

Image 3: Images of our workstation during development of our cultural probe (III)

Deploying the cultural probe

As mentioned, our users had the probe kit for the duration of four weeks. As preparation for these weeks, the users were introduced to the kit through the aforementioned introduction pages, as well as a conversation with one of us, where they could ask any clarifying questions before starting. Thereafter, it was up to the users to engage with the kit as they wished. In order to keep track of their progress we scheduled touchpoints with the users once a week. Some of these touchpoints were over a phone call, others were over text. For the touchpoints we asked questions regarding their progress such as:

- How has it been so far?
- What is your general experience of the exercises and the kit so far?
- Have you made any changes to the tasks so far - and what changes?

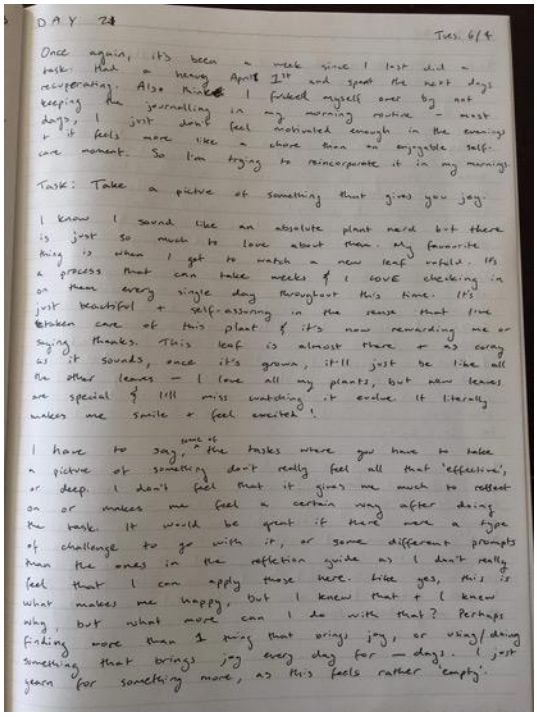
The questions varied from time to time, depending on the user's engagement. During these touchpoints, we were able to gain insight of what our data might consist of. More importantly, we served as friendly reminders for the users to keep engaging with the kit.

Another way that we could keep track of our users' engagement was through the image tasks. Here, we would continually receive images through the snapchat account, which made us sure that the users were in fact engaging with the probe kit.

Image 4: Simone writing to her future self.

Image 5: Eva sending an image showing something she use when upset.

Image 6: Mette sending an image showing something she would like to improve upon.



DAY 24 Tues. 6/4

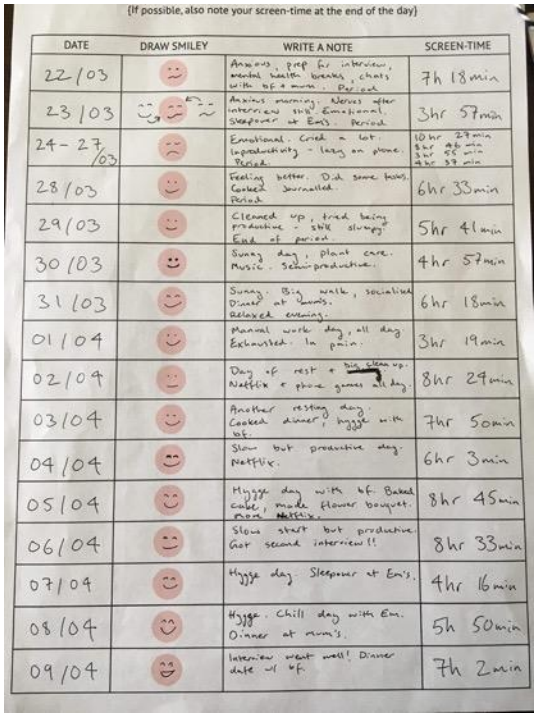
Once again, it's been a week since I last did a task that a heavy April 1st and spent the next days recuperating. Also think I fricked myself over by not keeping the journalling in my morning routine – most days, I just don't feel motivated enough in the evenings + it feels more like a chore than an enjoyable self-care moment. So I'm trying to incorporate it in my mornings.

Task: Take a picture of something that gives you joy.

I know I sound like an absolute plant nerd but there is just so much to love about them. My favourite thing is when I get to watch a new leaf unfold. It's a process that can take weeks & I LOVE checking in on them every single day throughout this time. It's just beautiful + self-assuring in the sense that I've taken care of this plant & it's now rewarding me or saying thanks. This leaf is almost there + as crazy as it sounds, once it's grown, it'll just be like all the other leaves – I love all my plants, but new leaves are special & I'll miss watching it evolve. It literally makes me smile + feel excited!

I have to say, some of the tasks where you have to take a picture of something don't really feel all that 'effective', or deep. I don't feel that it gives me much to reflect on or makes me feel a certain way after doing the task. It would be great if there were a type of challenge to go with it, or some different prompts than the ones in the reflection guide as I don't really feel that I can apply those here. Like yes, this is what makes me happy, but I knew that + I knew why, but what more can I do with that? Perhaps finding more than 1 thing that brings joy, or using/doing something that brings joy every day for – days. I just yearn for something more, as this feels rather 'empty'.

Image 7: An example of a journal entry.



(If possible, also note your screen-time at the end of the day)

DATE	DRAW SMILEY	WRITE A NOTE	SCREEN-TIME
22/03		Anxious, prep for interview, mental health breaks, chats with bf + mum. Period	7h 18min
23/03		Anxious morning. Nerves after interview shit. Emotional. Sleepover at Em's. Period	3hr 57min
24-27/03		Emotional. Cried a lot. Unproductivity - lazy on phone. Period.	10hr 29min 5hr 46min 3hr 55min 4hr 38min
28/03		Feeling better. Did some hobs. Cooked. Journalled. Period	6hr 33min
29/03		Cleaned up, tried being productive - shit sluggy. End of period	5hr 41min
30/03		Sunny day, plant care. Music. Semi-productive.	4hr 57min
31/03		Sunny. Big walk, socialising. Dinner at Laura's. Relaxed evening.	6hr 18min
01/04		Manual work day, all day. Exhausted. In pain.	3hr 19min
02/04		Day of rest + big clean up. Netflix + phone games all day.	8hr 29min
03/04		Another resting day. Cooked dinner, hygge with bf.	7hr 50min
04/04		Slow but productive day. Netflix.	6hr 3min
05/04		Hygge day with bf. Baked cake, made flower bouquet. mom. Netflix.	8hr 45min
06/04		Slow start but productive. Got second interview!!	8hr 33min
07/04		Hygge day. Sleepover at Em's.	4hr 16min
08/04		Hygge. Chill day with Em. Dinner at mum's.	5h 50min
09/04		Interview went well! Dinner date w/ bf.	7h 2min

Image 8: An example of a filled out mood journal.

In general, we did not want to interfere too much with the users during the four-week period, as we did not want to influence their behaviour, with our input or possible micro-management. At the same time, we were dependent on the data that would be produced through the kit. Therefore, we had to ensure that the users were engaging with the kit, otherwise we would have had to engage with an alternate plan for empirical work during those four weeks. The data we received from our users consisted of their mood tracker, images from the snapchat exercise and pages of their journal with their reflections (image 4-8). The users wished to keep the entire probe kit for future use.

Evaluating the cultural probe

After the four-week period we decided to host a participatory workshop with our five users. The workshop had a dual purpose for our empirical work. First, we wanted to have a discussion about the context in which we are working. This context was about low self-esteem and what coping mechanisms we use to combat low self-esteem. We wanted the users' perspectives on this in a workshop, as we had an assumption that they would have reflected upon this during the four-week period. Secondly, we wanted to evaluate the probe kit and the exercises that the users' had gone through. The evaluation of the probe kit was done collaboratively with the users, as we believed that they would be able to validate each other's experiences and assumptions through an

oral discussion. Before the workshop, the users had sent their reflections and images back to us, so we could quickly review their data before the workshop - in this way we were able to ask specific questions about experiences, that the users might have had in common, based on the reflections they had shared with us during the workshop.

The workshop was held online in a Zoom meeting on April 11th, where all 5 users were able to participate. The workshop lasted two hours, and a full transcription can be found in appendix C. The online workshop and collaboration tool Miro was used to host the workshop. Here we had prepared several boards (appendix D), where each represented a task designed for the workshop. Here, each user could find digital post-it notes as well as dots (for dot-voting) under their own name.

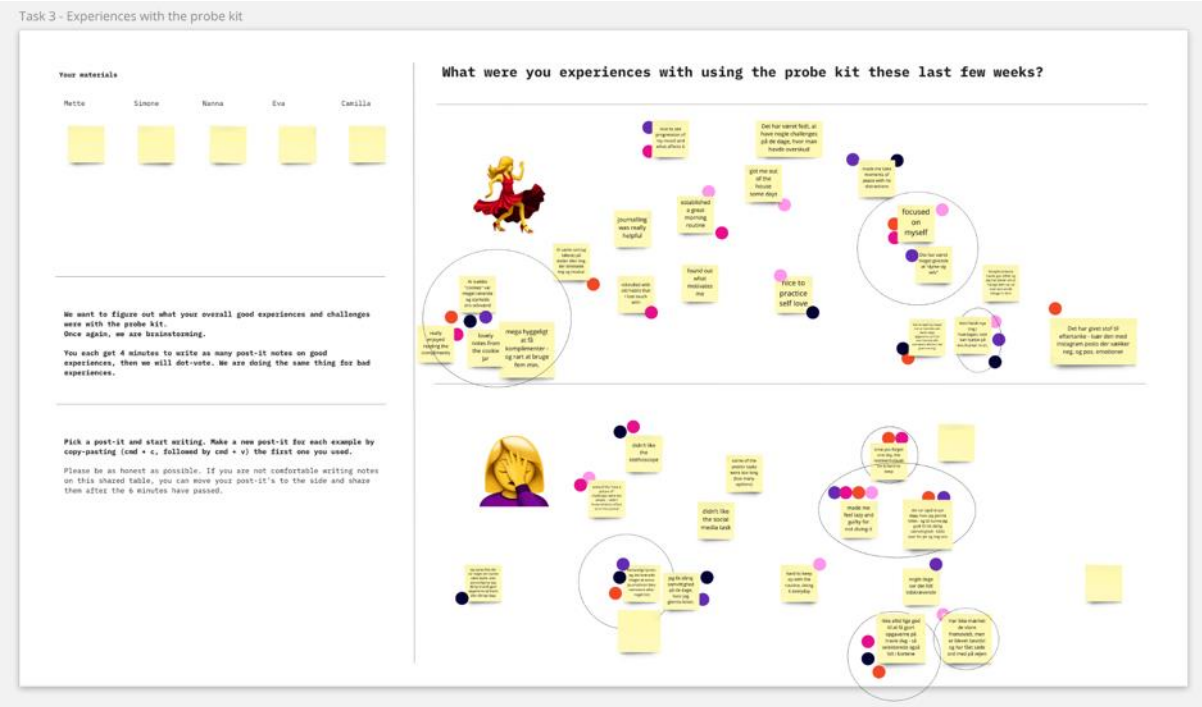


Image 9: Example of a miro-board used in our participatory workshop.

As the workshop had to be held online, we wanted to seek inspiration as to how online workshops are best structured, in order to create a space where participants feel that they can express their opinion. It is known to be difficult to engage a user in an online exercise, when they are isolated physically (Gillett-Swan, 2017). Here, it is important to create a very structured program, where the users know exactly what to do, and also feel safe communicating. We had experience using the Miroverse for inspiration on workshop exercises and here we found the Lightning-Decision-Jam (LDJ) created

by the German digital design agency AJ&Smart (Miroverse, n.d.). LDJ workshops are good for investigating broad topics, where you need to create a concise understanding of said topic in a short amount of time (max. 2 hours). The most used methods during this workshop was brainstorming and dot-voting. We also used a method called effect vs. effort, which is a graph where the participants help placing certain activities based on the effect of the activity and the effort that must be put into the completion of the activity.

It was also planned to use the How Might We (HMW) method as part of the evaluation of the probe kit. Here, we learned that this method is very grounded in design work and should have a more thorough explanation before being applied to a workshop with users, who are not used to doing this type of workshop. Further, the HMW method demanded that the users had a more elaborate knowledge of the research field in order for them to see solutions to the challenges that they had previously identified with the probe kit. After working our way through this exercise with our participants, we decided to end the workshop, as we felt that it would become tiresome for the participants to continue further into a more elaborate *Making it real* exercise. We also saw that we ourselves could complete the rest of the activities, and it might also be more beneficial to do so, considering our experience with design work. We concluded that we had already gained the insights needed about the experiences with probe kits and the core areas that are important to design for, when designing for young women experiencing low self-esteem.

Working with empirical data

After our workshop, we started familiarising ourselves with our dataset. This was done through the previously described method, coding. We had several iterations of coding starting in sorting data entries into sub-categories of tasks and design elements in the probe kit and ending in codes based on emotional reactions and experiences with the probe kit.

A snapshot into our process of coding can be seen on a snapshot from the Miro board, where we were able to collaboratively code all the data online (images 10-11).

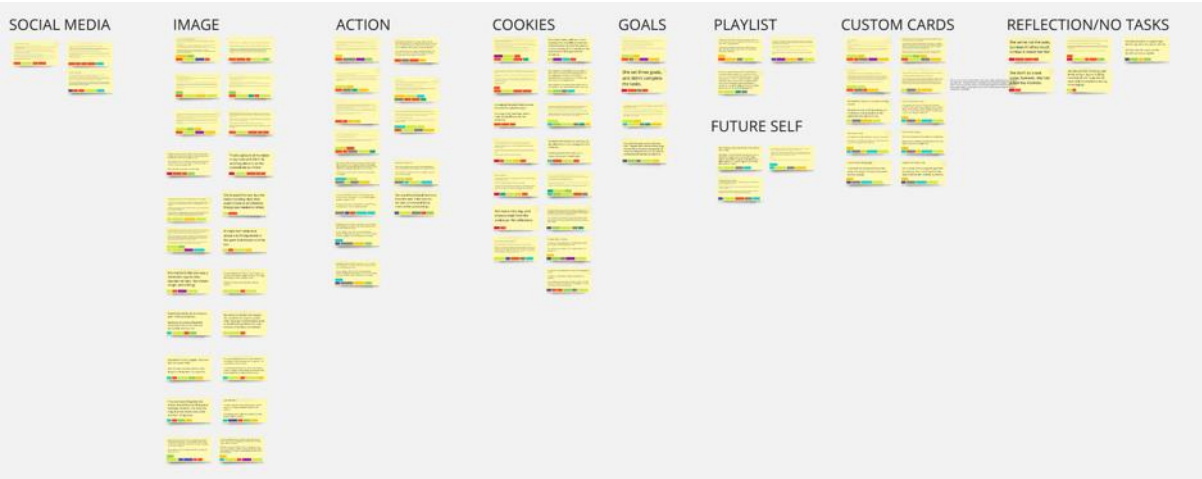


Image 10: Screenshot of a sorted view of our initial coding.

Image 11: Post-it's with several codes as a result of several iterations of coding.

The post-it's made in the Miro board, are summarised memos of data entries that we found relevant to our project in order to contribute to answering our research question. Some parts of the reflection journals, mood trackers and images from the probe kit have not been actively included, as they worked more as personal reflections on private affairs rather than reflections relevant in building routines around self-appreciation. Our codes worked as the foundation for our analysis and creation of design arguments that will be presented in the next chapter.

Findings and Analysis

In this chapter, we will be presenting our findings from our probe kit and workshop data. The data from the cultural probe consists of journal entries, in which the users reflected over the tasks from the kit. Further, we will include findings from our workshop, simultaneously citing from the workshop transcription, as well as snapshots from the Miro boards that our users engaged with during the workshop. Our findings will be structured through categories and codes found through our analytical process with coding of our empirical data. Here, we have identified three overarching categories: *consciousness raising, self-appreciation* and *building routines*. These categories will help us answer our research question, while also arguing for design attributes that will be embodied through a design workbook presented after this chapter.

Consciousness raising

The first category is called consciousness raising. This term is based on the Transtheoretical Model. We have borrowed this term, as we have seen in our data, that the code ‘becoming aware’ and its sub-code ‘reflection’ were some of the most frequently occurring codes in our dataset. In summary, this term of consciousness raising, coined by Prochaska and Velicer (1997), is about becoming aware of the causes, consequences, and solutions to problems. Interventions that instigate consciousness can include education, feedback, confrontation and interpretation. We have learnt that becoming aware of the causes and consequences of one's actions (both good and bad) can help in building more positive routines for better self-esteem.

Becoming aware

Becoming aware is a code used for all data entries where a user is experiencing a realisation of having bad or good habits that they either suffer or benefit from. Before an individual can make positive changes, they must become aware of positive and negative behaviours that have an impact on what they wish to improve upon (Prochaska and Velicer, 1997). Becoming aware is one of the most important stepping stones in building positive habits for self-esteem. Users become aware of negative behaviours, things they would like to improve or things that they forget to appreciate about themselves. This makes them feel more motivated and open to make changes.

In our workshop, we found that becoming aware of negative behaviours was vital to improving one's routines for self-appreciation. While defining low self-esteem collaboratively in the workshop, our users initially identified that destructive thought patterns were a big contributor to low self-esteem. These thought patterns were identified as over-analysing experiences, spiralling, doubting oneself and being overly critical to one's own actions (image 12).

Image 12: Screenshot of post-it's concerned with negative thought patterns.

During the period with the probe kit, the users had become aware of negative thought patterns through the following exercises: creating a list of things that make them happy, positive affirmations, finding reasons behind the pattern of thought and replacing negative thoughts with positive ones. (image 13)

Image 13: Screenshot of post-it's concerned with becoming aware.

In data from our probe kit we saw several journal entries, where users became aware. First, a user came to the realisation that she feels more unhappy than happy. Nanna explains that she has not previously reflected upon this, but through the exercises in the kit she has become aware that this might be an issue. Further, she explains that this could be the reason why 'small things' often upset her: "*I don't feel any different in my well-being, but I have noticed that I am more unhappy than happy. Even though I have great days, I get upset by the small things*." (Nanna, Journal Entry #9)

Becoming aware of this is the first step in building positive routines for strengthening her self-esteem. Further, Nanna does not herself believe that she has made any progress, but this is where we argue that the small realisations or moments of becoming aware of destructive patterns, are the actual changes that we should be aiming for.

Another instance of becoming aware is also from a journal entry reflecting on the task 'Send a text to your future self'. Here, Simone doesn't benefit from looking to the future, as it sets expectations for herself. Rather, she finds comfort in the idea of speaking to her past-self, and telling her about things that she has already accomplished: "*The future 'scares' me a little, and because I wrote a thing that I wish I would change for the future*" (Simone, Journal Entry #1)

In this entry, Simone becomes aware of things that she has accomplished rather than things she will/could accomplish. It serves as the foundation in becoming thankful for things that she already has. The awareness is about how she should set goals for herself, that are approachable here and now, because she will experience guilt for not reaching a goal that she would set for a future version of herself.

As described earlier, some exercises were created in order for users to reflect upon their social media usage. As part of this, there were a few journal entries concerning 'becoming aware' of the negative ways in which some users would engage with social media. For instance, one user realises how she has often posted pictures from a nice situation, at times where she felt low, in order to feel better about herself. When she looks back at the picture, she doesn't reminisce about the good thing in the picture,

but the state of mind she was in, when posting it: "*A little thought provoking that it is the images of myself, that I have posted in periods where I felt worse than when the picture was taken - guess I needed confirmation*." (Simone, Journal Entry, translated, #14)

Social media is used to gain quick and easy positive feedback. It is very easy to stage an alternative/better reality through social media. This was prompted by a task, where she had to find two Instagram posts that affected her positively and two that affected her negatively. Another user moved further in her realisation. Here, she had identified a negative behaviour regarding social media usage described in the quote below, and decided not to use it for a period of time in order to avoid the distraction and time consumption that is often part of social media use.

> "*I do waste a lot of time there and I have an article due tomorrow so I will be using this as an opportunity to be productive with minimal distractions, and also to be present with my mum. (...) I instead played a quick game (puzzles) and got back to work much faster - usually I'd be swallowed in by the 'autoplay' feature on Facebook and waste a lot more time, sometimes without realising it."* (Camilla, Journal Entry #10)

If we look back at the Transtheoretical Model, we see that the behaviour that a user wants to change is often concerned with some type of negative behaviour, such as addictive or abusive behaviours. Here, Andreassen & Pallesen (2014) argue that SoMe use can become so uncontrollable to the point where it becomes addictive. They define SoMe addiction as "*being overly concerned about social media, driven by an uncontrollable motivation to log on to or use social media, and devoting so much time and effort to social media that it impairs other important life areas*" (Andreassen & Pallesen, 2014, p. 4054). For this exact reason, we wanted to have at least one exercise that prompted the users to reflect on their social media use, which in turn might lead to becoming aware. In our evaluative workshop (image 14) we also saw that users in fact did come to the realisation that what might help build better self-esteem would be to 'switch off social media' or at least be more considerate about the profiles they were following.

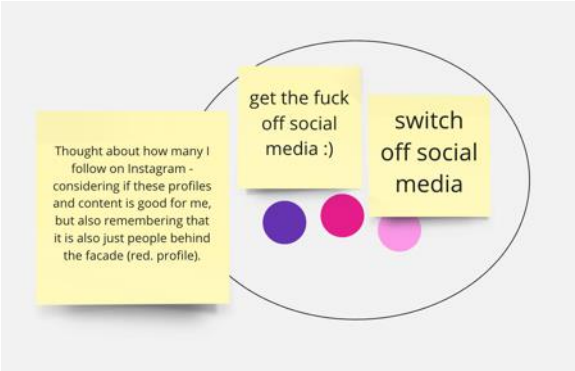


Image 14: Screenshot of post-it's concerned with Social Media.

Reflection

A sub-code to 'becoming aware' is the code 'reflection'. Reflections are a key part of changing behaviour. Before 'becoming aware' the users must reflect on how certain actions affect them. In this project, we used a journal as a tool for reflection. It was subjective from user to user how they preferred to reflect upon a task, but one user notes that she found reflection through journaling to be very beneficial.

> "*I've confronted internal thoughts and even managed to do something about some of them; it has given me a reason to take a break from a screen and just sit, ponder, write and enjoy the moment."*

She continues in a later passage about a job interview;

> "*I'm hoping that once I write down how nervous I am, how scared I am that I won't know how to answer questions, how excited I would be to get this job and how disappointed I would be if I didn't, it'll help me discard those worries from my mind and into paper"* (Camilla, Journal Entry #16)

This is a good example of how a user used reflection actively to become aware of negative habits, and set goals to change them. The user notices that the externalization of her thoughts from mind to paper helped her '*manage to do something about some of them'*. This is where reflection can help a user change perspective and learn how to handle destructive thought patterns. Further, we see that

destructive thought patterns mainly become destructive when they stay internalized (i.e. are not discussed with others or written down for later reflection). When considering the term self-liberation (Prochaska and Velicer, 1997) concerning believing in one's own capabilities for change, we believe that this type of externalisation of destructive thought patterns can help users set more realistic and in-the-moment goals to change these patterns. Setting goals is one of the methods for self-liberation.

In regard to reflection, we have also noted that in several reflections, the users seem to reach a somewhat meditative stage. This is identified through the three axioms of mindfulness (Shapiro et al., 2006). The users set an intention to write in the journal and focus their attention inward. Further, the users work with their attitude towards writing about their experiences of self-appreciation throughout the time they spend writing in the journal. The journaling itself is not to be understood as the meditative practice, but rather as a support in meditative practices, prompted by the cards. Some users found it difficult to experience meditative states, but the action of reflection made them set an intention for the experience (or the next experience), reflect on their attention (did they manage to focus on the task why/why not) and reflect on the attitude that they started the exercise with. These reflections helped them in their work with the upcoming tasks.

In summary, we found that users often become aware when using the reflection journal. Here, the reflection journal and reflection guide was helpful. Further, it was subjective to each user which tasks prompted becoming aware of negative thought patterns. An important finding here, is the fact that it is subjective to each individual what will prompt the initiation of a process of exercising self-appreciation. Further, a finding is that becoming aware is a first step to figuring out what actions of self-appreciation are valuable to the individual. In a future design, we argue that there must be elements that make the user reflect in order to become aware of negative thought patterns.

Self-appreciation

Our second category is that of self-appreciation. Self-appreciation encompasses how we perceive ourselves, and our ability to assess our self-image in a positive light. Self-esteem, as defined in the first chapter, concerns how we evaluate ourselves. The ability to reassess the way in which we see ourselves, is crucial in building self-esteem. This can be linked to the elements in the transtheoretical model of behaviour change (Prochaska and Velicer, 1997) of self-reevaluation and self-liberation that concern a person's self-image and perception of a person's own capabilities. In building behaviour changes, a person not only has to become aware, but also adapt changes to their mindset.

When individuals are under 'self-threat', essentially when individuals feel that their self-esteem is at risk. For example, when we experience failure or rejection, some tend to seek affirmation and support from others, more likely those they care about, as they can provide them with a sense of social acceptance (Park & Maner, 2009). However, seeking acceptance from others is also accompanied by risk; if they don't find acceptance, it can be a source of rejection and further negative evaluation, rather than a positive support (Park & Maner, 2009). We mention this, as although seeking support from those around us is important, being dependent on the words of others is a risk to worsen our self-esteem. Therefore, also why working with self-esteem and our sense of self is important, we need to learn how to be able to seek acceptance within ourselves. This is also in line with mindfulness practices that have the goal of being compassionate to oneself.

The category of self-appreciation, in the context of our data, is about tendencies that we have when regarding our own self, as well as techniques in how to address the challenges in self-reevaluating (Prochaska and Velicer, 1997) our self-image.

Comparison

The first code is comparison, and what is meant by this, is the human tendency to compare ourselves to others, in a way that negates our own value. This is often substantiated by our predisposition to setting unrealistic expectations to ourselves, whether it be around our appearance, performance or experiences.

Comparison is most often an internal debate that isn't necessarily substantiated by legitimate claims. We are able to create and set expectations to ourselves that other people wouldn't have. Therefore, what is acceptable for others, is entirely different from what we consider acceptable for ourselves. This is also what we have previously understood through the looking glass-self (Cooley, 1983). The looking glass-self is a metaphor for how individuals will use an imagined person's judgement to pass judgement over themselves. The three components to this metaphor are: the imagined appearance to others, the imagination of others' judgement and the feeling of pride or mortification based on this judgement.

This was especially clear in a journal entry where one of our users' self-perception versus how others saw her, were entirely different. She herself, was experiencing that everyone around her was reaching their goals in life, and was unable to recognise that in her own life. In one of the cookie-jar tasks, Mette read a message that said "*You work hard to reach goals that you have set for yourself, and you usually reach those goals*" (Mette, Journal Entry #15, translated). She expresses in the journal entry, how this made her cry. In her own words, she reflects on how this is something that she constantly reflects on; that she sets goals and doesn't feel like she achieves those goals.

In the cookie message, a significant other recognised that she does indeed set and achieve goals in life, which can be attributed to her strong reaction to the message. Her own self-perception is dissimilar to how others see her. Further, in her reflection of this task, she expresses how she compares her own progress to others, and that is part of what prompts her dissatisfaction with her own problems. She only sees others achieving their goals, but doesn't acknowledge when she herself achieves goals. This in turn resembles the three components of the looking glass-self metaphor, where Mette imagines how others might view her progress in life, while passing judgement on herself based on this imagination. Luckily, in this instance where she reads the cookie message, she feels pride, as she realises that this imagined judgement is untrue.

The idea of how comparison influences our self-esteem, was heavily addressed and discussed in our workshop. It became evident that comparison is a major trigger and contributor to feelings of low self-esteem. What was especially interesting, was the fact that all of our users acknowledged this as a problem, but also an irrational problem. They all agreed that comparing ourselves to others prompts unfair judgments on ourselves, and that these comparisons are completely internal and most often not based on any factual evidence, but rather based on the imagination of others' judgement and the feeling of mortification based on this judgement (Cooley, 1983). So although they all acknowledge that comparisons aren't legitimate, they still see and know the extent to which it affects their self-esteem, however irrational.

In our workshop, our users explained the idea of comparison in that of exercising self-critique. Our users came forth with the idea of comparison when discussing "What is low-self-esteem to you?". Many definitions concerned the idea of comparison with post-it's such as "critiquing everything you do, but when other people do it, it's normal or fine" or "feeling that others are/look better than you" and especially thinking of how others perceive and think of yourself (image 15).

Image 15: Screenshot of post-it's concerned with comparison.

It seems that the way we think others see us has a huge influence on how we see ourselves. What is important here, is the idea of what we think others see, thereby acknowledging that this is not necessarily a truth but merely a feeling that we have created in our own minds as the looking glass-self metaphor addresses. Emler (2001) also explains that it is not so much the evaluation from objective others that is harmful to self-esteem, as much as it is a harsh and unrealistic self-evaluation that is often based in unrealistically high ideals.

Furthermore, Social Media has and does play a huge part in our self-evaluation. Social media doesn't only provide a platform for self-presentation, but it provides us with a plethora of people and personas to compare ourselves to. This in turn, allows for individuals to create certain expectations for themselves, on how to look, what to be doing and how to be. For example, Camilla expressed her own experiences and opinions on social media through an Activity task where she had to log off of social media for 24 hours. In this, she expressed how she had previously removed herself from the platform of Instagram, because of the social pressures that she felt came along with the platform. However, more importantly, she emphasises on how Instagram is a platform designed for comparison: "[Instagram] *leaves you with a sense that your life needs to be up to par with others. It makes you doubt yourself, but also doubt your friends and who you thought they were*." (Camilla, Journal Entry, #17)

Comparison is recognised as being irrational by our users, however it still has a great impact on their self-esteem. Comparison is experienced through the people they see and meet on a daily basis, and further complicated by social media that allows for comparison in a moment's notice. We believe that by making young women feel more secure in their 'self', they will be less likely to find their worth through comparing themselves to others. In future design work for assisting in practices of self-appreciation, we emphasise the importance of helping users in using comparison in a way that will have a more positive effect on their self-esteem.
We also see value in moving away from the smartphone in the design of digital technologies on this matter, as this can allow for distance between the user and the negative influence that social media can have when working with self-perception.

Guilt

We speak of guilt in the sense of the negative emotion that comes from the experience of not living up to our own or others expectations. Guilt was a code we primarily determined from a discussion in our workshop, of which we discussed negative aspects with the probe kit. Our users expressed feeling guilty when they didn't carry out the tasks or forgot to use the kit (image 16). In our onboarding of the kit, we emphasised that the kit was to be used at their own pace, in a way that they felt comfortable, even if this meant to skip days of using the kit. Although we expressed this, with the intention of removing guilt, it didn't entirely work. This could possibly be

due to expectations that they had set for themselves, rather than expectations that we had set for them.

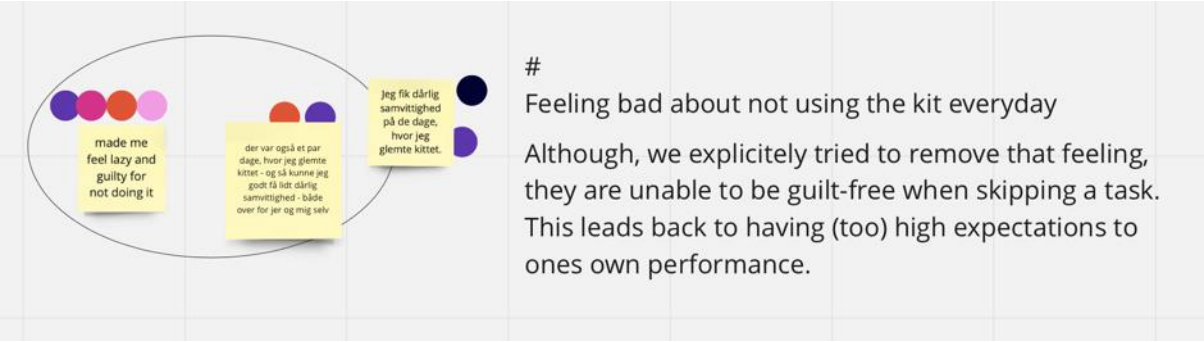


Image 16: Screenshot of post-it's concerned with guilt.

Guilt is an emotion that often is accompanied with feeling disappointed and frustrated with one's own actions. This can also be set in comparison to the Transtheoretical Model's stage on maintenance, in which many experience 'relapses' in the development of behavioural changes (Prochaska and Velicer, 1997). Specifically, as presented in our probe kit, our users feel guilty in the context of 'relapsing'. When our users don't live up to expectations that they have set for themselves, they feel guilt, which in turn affects their self-esteem. In relation to self-appreciation, feelings of guilt are relevant in the context of treating ourselves with compassion. Rather than feeling negative feelings of guilt and self-blame, user's should rather accept actions and appreciate traits within themselves, as part of building their self-esteem.

Emler (2001) further elaborates on feelings of failures and successes, as crucial contributors to why self-esteem can fluctuate. When a person experiences success, their self-esteem can rise, and feelings of failure prompts self-esteem to lower. Guilt is a feeling that accompanies experiences of failure, and what a person considers a failure is most often self-determined. For example, our users had set goals for themselves in using the probe kit and carrying out the activities. When users missed days of activities or didn't use the kit as much as they had thought, they felt guilt, e.g. Eva expressed in a reflection "*I feel guilty for not having been better to draw tasks and writing in my journal*" (Eva, Journal Entry, #3, translated) and Nanna also expressed feelings of guilt whenever she would forget to do the tasks.

In addition, our users also expressed that some of the tasks prompted feelings of guilt. Especially the image task of: “Take an image of a place that makes you feel stressful” (image 17), was triggering for a majority of the users.

Image 17: An image card being carried out by a user by using Snapchat.

For example, Nanna expressed in her journal entry about the task: “*I took a picture of my clutter in my room and didn’t do anything about it, so that stressed me out more*” (Nanna, Journal Entry, #3). Simone also expressed her feelings on the task, “*It stresses me that I should clean [the mess] and sell my things, but I don't get it done*” (Simone, Journal Entry, #5). It seemed that it was rather that she didn’t clean the mess, i.e. the guilt, that was a stressor than the actual mess. For some users, the image task prompted an action of cleaning the mess, but for others it served as a guilt-fuelled reminder.

Guilt is driven by the expectations that we set for ourselves, or that others set for us. In our design space, our aim should be to create a guilt-free space, by emphasising that building self-esteem is about building habits, and that is a process that not only requires time, but patience. As the Transtheoretical Model presents, when aiming for behaviour change, we will experience set-backs, and it is within these setbacks, that an individual has to address feelings of guilt, stress and frustration, on their relapse, with compassion, patience and understanding. They have to instead accept it as a part of the process. In our workbook we will explore ways in which we can assist users in accepting feelings of guilt as part of said process. We believe that it is important to

address guilt, as a natural part of behaviour change, because this is where users might feel discouraged enough to end their work towards that change in behaviour.

Affirmation

When working with self-appreciation, the simplest form is to bring the same emotional support and encouragement that we give to and desire from, others, to ourselves. When discussing the code affirmation, this term is understood as the acknowledgement of our positive traits within ourselves, and the ability of the user to see and recognise their own value. As previously explained, our global self-esteem builds on three elements: performance-, social- and physical self-esteem (Rosenberg et al., 1995). It is subjective to each individual, of which elements that we gain and/or lose our self-esteem from. Some might find that their self-esteem is mostly at risk in regard to their physical appearance, where others are pressured in social situations. The custom cards and variety among the tasks were primarily motivated by this factor, that each person is individual and unique, and that each user needs something different when working with affirmations.

A reason why some find self-esteem to be difficult to improve, is the inability to see value within ourselves that is obvious and clear to those around us. As expressed through our comparison code, self-esteem is driven by emotion, rather than logic and rationale. One of our user's was confronted with this idea, when reading a compliment from the cookie jar task that expressed "You are really good at noticing when others are feeling low, and help to your best ability": "*Sometimes I can feel that I should help my friends even more or be better at seeing signals/asking them questions, so it was nice that others could recognise that trait within me*" (Simone, Journal Entry, #3, translated). In this journal entry, we experience a user's reaction when confronted with a compliment that specifically addresses an insecurity that she had. The cookie task was accompanied by the task of having to retell the compliment to herself repeatedly, while listening to her own heartbeat. She reflects on the experience with the following: "*In the beginning I felt like I was 'lying' when I was thinking it, but towards the end, it became better and afterwards I felt an urge to call a friend and hear how she was doing."* (Simone, Journal Entry, #3, translated). What began as a 'lie' turned into an affirmation that the compliment was truthful; she became aware that her self-image might not match reality. Moreover, Eva experienced through the same exercise, that

the affirmations that she received from her significant others, were important and meant a lot, as they were things that she was struggling to telling herself: "*I drew some really sweet cookies, that affected me deeply and it was really encouraging, in relation to my self-confidence for my exam, to read that my friends had this view of me, that they wrote*" (Eva, Journal Entry, #14, translated).

In general, as also expressed in our workshop, the users found that the compliments were nice and they appreciated them, especially as they came from people they cared about. The users expressed that they would hold on to the compliments, and some of the users also considered using them in the future, for when they would need a boost to their self-esteem. (image 18).

Image 18: Screenshot of post-it's concerned with experiences with the cookie jar.

Self-affirmations are in theory, low in effort. The act in itself, is simple - it merely requires a willingness to tell ourselves that we have worth. However, although the act is simple, it is not easy to do. Self-affirmations are often about reprogramming the brain, i.e. telling yourself something that you don't necessarily believe, until you finally believe it. We saw that through Paay et al. (2018) Happy Bits, that giving ourselves compliments can be difficult, so we drew on compliments from significant people in their lives. The idea was, that if they received compliments from those they cared about, they might begin believing those things and therefore adopting that self-belief into their own routines. This can be paralleled to the Transtheoretical Model, where a

person might begin to prepare for change (Prochaska and Velicer, 1997). We see this with one of our users. As a reaction to the cookie task, Mette reflected on the low effort and high reward incorporating affirmations into a routine could have. She reminded herself that compliments are important and they have a substantial effect. The compliments from others, made her contemplate on further wanting to implement it in her life by being kinder to herself, as she states: "*I will probably think more about giving myself a few nicer words along the way, after this task*" (Mette, Journal Entry, #15, translated). This user indicates to us that she is experiencing what Prochaska and Velicer (1997) coined as consciousness raising, in that she has acknowledged the value of compliments from others, reflected over its impact and relatively low effort, and adapted it to be something she can do for herself, independent of others.

Based on our data, we see that a users' self-image, in most cases, doesn't match how their significant others perceive them. In these cases, affirmations from others prompted self-affirmations and were a way in which they could re-adjust their negative self-perceptions.

Mindfulness

Earlier, we presented the three axioms of mindfulness that both explain how to practice it, what it means, and essentially why it works. To reiterate, these axioms were: attention, intention and attitude. Mindfulness has been proven to be effective in helping with low self-esteem, primarily in its ability to teach individuals to be compassionate and empathetic towards oneself, as expressed in the attitude axiom. We incorporated aspects of mindfulness through various tasks in the probe kit, with the purpose of seeing what type of reflections it would prompt from our users. We found that most of our users had a positive mindset towards mindfulness, and were excited by tasks that focused on creating a situation for mindfulness. Our users weren't told about the specifics of mindfulness, but were able to reflect on activities and actions that we recognise as being mindful. For example, one of our users, reflected in vivid detail on one of her walks, as part of an image task prompting her to take a picture of a place that makes you feel calm.

> "*I get a sense of calm, when I look over glittering water on sunny days. I therefore chose to head down to Århus Ø. I could feel that the fresh air and a*

> *little exercise, got my thoughts in order - small speculations, worries, over-thinking, became more solution-oriented and rational thoughts*." (Eva, Journal Entry, #6, translated).

These go hand-in-hand with principles and practices of mindfulness. The tasks in itself, and her willingness to make an effort with the task, is comparable to the axiom of intention, where she sets a goal for herself. She further explains in her reflection, reasons for enjoying this task: "*I enjoyed this task, as it made me place words on to what actually makes me feel comfortable*." (Eva, Journal Entry, #6, translated); this can be compared to the principle of attention, in that she brings focus to her thoughts. Further, her attitude is compassionate, and she is understanding and reflective on her needs: "*It works for me, to know the little things that I can do well for myself - especially in situations, where I feel down and am more inclined to stay in bed. It can be a good source of energy, but also be the exact opposite*." (Eva, Journal Entry, #6, translated).

Similarly, Mette acknowledged her tendencies to fill her calendar with things to do, in fear of boredom. She further reflected on the value of taking a break and finding peace, which the image task prompted: "[I] *forget to take a step back and breathe, just for myself*." (Mette, Journal Entry, #13, translated).

Further, one of our users, Camilla, in one of her custom cards, tasked herself to do crosswords/sudokus, as it is something she enjoys, but has forgotten to give time to. Through this activity, she found time to be mindful, as she reflects it "*Makes me happy and lets me be more aware of my surroundings"* (Camilla, Journal Entry, #21). What was interesting, was that this user directly made a comparison to her smartphone, as an anti-mindful device:

> "*I think if I were, say, in the sun on my phone, I wouldn't be as grateful. When I'm doing something else [than using a phone], there are natural breaks when I look up, look around, feel the sun on my face and can actually appreciate the moment.*".

She further reflects on the smartphone, and its effect on her/individuals:

"*Time moves slower when you're not absorbed by your phone*" (Camilla, Journal Entry, #21).

Like with Kuhn et al. (2019) research on mindful interactions with smartphones, they found that spending more time 'away' from the device, i.e. by making interactions with the device more difficult, allowed their users to avoid the compulsiveness of the smartphone. Our user, Camilla, is essentially criticising the non-essential parts of a smartphone, or more specifically social media, like the never-ending stream of content that are designed in platforms for disengaging a user from the real world, by using designed content that engages users for hours.

What we can see is that mindfulness is an effective tool, and can be practiced and adapted to an individual's needs in a way that is easy and approachable. Mindful activities allow users to clear their minds from distractions and centre the user, so they can better focus on things that fill them with positivity. The importance in mindful activities is that they shouldn't seem alien to the users. We see that traditional mindfulness practices are often associated with meditation, which isn't necessarily accessible to all. For example, our users expressed difficulties with tasks that were too 'meditative': "*I've never been good at anything meditative - I am usually distracted or just not interested in clearing my mind/sitting still*" (Camilla, Journal Entry, #3). This is interesting, as the same user expressed earlier, her appreciation for enjoying the moment, without distractions. The idea that an exercise seems similar to that of meditative practices, can be intimidating for users. As our users engaged with the kit and were asked (indirectly) to practice mindfulness in new ways, they realised that they can practice mindfulness through everyday experiences such as; cleaning plants, reading, taking a moment of silence, watching the sunset, focusing on a compliment from friends, going for a walk or exercising. In our design, we want to normalise mindfulness and make it more attainable and easy for users to identify and incorporate into their own everyday practices.

Building Routines

Our final category is 'Building routines'. With this category we look into the foundational work that a user must do in order to build a new routine to assist them in being more

self-appreciative. Through our initial research and later in our empirical data we found that building routines is a big part of assisting young women in practicing self-appreciation and improving their self-esteem and happiness long term. Early on, we found that assisting in practicing self-appreciation should be done through the encouragement of behaviour change. Here, we have found inspiration in the Transtheoretical Model (Prochaska and Velicer, 1997) and the understanding of what it takes to change a behaviour that is deeply rooted in our habitual practices. Our probe kit and exercises were designed with this in mind. Through our probe kit we have been able to gain a deeper understanding of how behaviours around self-esteem and self-appreciation come to life in a day-to-day context. Even more importantly, our users also seemed to gain a better understanding of their current habits and behaviours that they wished to change or had already started to change. This brings us to our first code in this category; habits.

Habits

A habit is a settled tendency or usual manner of behaviour; an acquired mode of behaviour that has become nearly or completely involuntary; a behaviour pattern acquired by frequent repetition or physiologic exposure that shows itself in regularity or increased facility of performance (Merriam-Webster, n.d.)

This code was used every time a data entry had an insight about habits or routines that our users would struggle with. These struggles could concern the upkeep of good habits or trying to unlearn bad habits. We found it important to keep track of this, as one of our goals was concerned with building routines: The cultural probe should assist the users in building routines around self-appreciation. Here, building routines is what can lead to a behaviour evolving to a habit, which is an acquired mode of behaviour that has become nearly involuntary. In our evaluative workshop, a major finding was that the probe kit in large helped users with building routines in everyday life. Some notes made here were ‘established a great morning routine, rekindled with old habits I lost touch with, found out what motivates me’ (appendix D).

Image 19: Screenshot of post-it's concerned with help in building routines.

As part of building new routines, the users would also have to be able to identify habits that were undesirable. An example of such an acknowledgement is Camilla, who reflects on her use of social media in a Journal Entry (#10):

> *"I didn't realise that it (ed. checking social media) has become such an automatic reflex of mine, because I truly don't feel like I am addicted to it or need to check it all the time, but apparently it has become such a frequent pass-time that it is ingrained in my routine.*" (Camilla, Journal Entry #10)

Here, Camilla points to the reflex of checking Facebook through her phone in the morning before getting out of bed. She further realises that this behaviour has become ingrained in my routine. Thereby, this behaviour has now been identified as an unwanted habit. Camilla goes on to delete the application from her phone. She goes on in a reflection about how she managed to stop checking her phone in the morning. "*I am happy that I'm not finding a compulsion to go on my phone - relieved really*." (Camilla, Journal Entry #10). Camilla was able to see things that she thought would be difficult 'de-habiting' as more doable than expected.

Another user starts building a new morning routine due to a particularly stressful application process for university. For one of her custom cards she wrote "Take a long shower, use good products and pluck your eyebrows." She used the task as preparation and motivation to address an application. After the success of doing it one day, she continued doing it for the rest of the application period (lasts a week). She

explained that this was "*a simple every-day thing, that still gives somewhat of a spa feeling.*" (Eva, Journal Entry #2, translated)

Similarly, another user describes how she uses self-care through home-wellness days as methods for feeling better about herself and to energise. And realises that this is a practice that she should integrate into a weekly routine:

> "*I forgot a couple of days (ed. to use the kit), but today I am going to do home-wellness. It ended up with a foot bath, nail polish, facials and a nice body oil. I am actually pretty good at doing things like this for myself and pretty often, and should definitely make it a weekly thing.*" (Simone, Journal Entry, #15, translated)

Through the probe kit, we realised that building new routines is tricky; it demands a surplus in energy, an awareness of what will have a positive impact and strategies for avoiding bad habits. Some users had pre-existing methods for dealing with lack of energy which they used as part of the probe kit, where they added these methods to the three custom cards. These custom cards were things that the users enjoyed doing. In particular, two users used physical activity as a very important method for creating a surplus in energy to help them stay motivated: "*I needed a boost of energy. (…) Now I am filled with energy the rest of the day and I just feel awesome! I hope I keep up swimming for the rest of my life*." (Mette, Journal Entry, #5, translated).

For others, this motivation was fuelled through journaling, crossword puzzles, being around friends, cooking or wellness routines, but we'll get into motivating action in a later section of this chapter. Initially we will look into the struggles of staying motivated about building new routines.

De-motivated

Demotivated is a code that covers when users didn't feel 'emotionally available' to carry out the tasks, whether it be due to stress, lack of energy or motivation. One of our users expressed that the tasks, although motivated in the beginning of the cultural probe, were beginning to lose motivation, and some of the tasks felt like a "*chore rather than an enjoyable self-care moment*" (Camilla, Journal entry, #2). However, on a

different day, the same user, when presented with an activity task, decided to place the card back in the deck, for another day where she felt more motivated. In this case, she saw the value of the card, but acknowledged that it wasn't the right day or circumstance to do that specific task.

Our users varied on their motivation for the different tasks. For example, Camilla found journaling as a tool for finding calm and relaxation, while another user, Eva, found that journaling required energy. Therefore, when she was having a day with less energy, she found it difficult to be reflective, although she expresses enjoyment for the task. Furthermore, we found that motivation to carry out tasks was also linked to the value of the task at hand. In order for them to be motivated, even on days with less energy, users must perceive value in the task at hand. This is what we have defined as the sub-code of "Difficult/shallow". Here, we investigate the particular tasks users found less impactful.

For example, Eva describes her experience with the stethoscope aspect of the cookie task:

> "*Some of the cookies I was touched by, but it didn't affect my pulse. Maybe if the cookies were from a boyfriend/girlfriend, it would have affected me differently? Or maybe I just didn't let the words sink in*" (Eva, Journal entry, #9, translated).

Rather than embracing that this wasn't the task for her, she questioned the experience, thinking that she could have succeeded if she had carried the task out differently. As a contrast, Camilla was more clear and self-assured in what worked for her, and demanded more from different tasks, rather than placing doubt on herself. As she expresses with an image task:

> "*I don't feel a certain way after doing a task. It would be great if there were a type of challenge to go with [the image task], or some different prompts than the ones in the reflection guide, as I don't feel like I can apply those here*." (Camilla, Journal entry, #2).

The criticism she makes here is that she doesn't feel stimulated by image tasks. Further, since she values using the reflection journal to have an impact, she didn't feel that the image tasks were applicable for proper reflections. She expresses the task as being shallow: "*I just yearn for something more, as this feels rather 'empty*" (Camilla, Journal entry, #2).

Camilla is an example of a user who recognises her needs and reflects upon them, for example in regard to the cookie exercise on the use of the stethoscope: "*I think for me, it's simply too personal and transgressive, so I think I'll stick to feeling my heartbeat instead."* (Camilla, Journal entry, #22). On the other hand, we have Nanna, who merely acknowledged that a task didn't impact her, without further reasoning: "*The tasks are nice, but I don't reflect upon them*." (Nanna, Journal entry, #6).

All individuals experience tasks and activities in different ways. Further, they are impacted differently for each task. Some users found value in simple and easy tasks, while others craved more demanding tasks, as they welcomed the challenge. This can partly be attributed to the fact that we are all different people with different needs. It could also be attributed to being in different stages of the behavioural change model. Camilla could represent a person who is in the action stage, as she has already made a change. Nanna's hesitance and difficulties with some of the tasks, could be explained in that she is just in the beginnings of the contemplation stage. In this stage, she might have an idea of making actionable changes, but it isn't an action she is willing to take in the moment.

We find that there are no specific tasks that most users like or dislike. This varies a lot from user to user. It is very subjective to individuals what will have an impact on their self-esteem. In a design for building habits for better self-esteem, it is important to let the users decide what they want to do, as they know best what will have a positive benefit on them. When that is said, we also acknowledge that it can be difficult to identify what might have an impact, if users do not already know. Therefore a design should also allow the user to explore what works for them, and give them the autonomy to choose what works for them, and what doesn't.

Motivate action

This code is concerned with actions that users take, beyond what the tasks prompted them to do. Here we have noticed somewhat of a snowballing effect, where one positive action towards changing behaviour can lead to further actions. When users have proven to themselves that they have the willpower to change some behaviours they might feel more motivated to keep up the change.

We found this to be valid with a few of our users. For instance, one of our users, experienced the positivity of one task merge into and manifest into other actions. The initial task was to go for a long walk. However, after the task, she felt the urge to reach out to a friend, because of the positive experience that she had from the walk. Similarly, the user went for a walk as part of a different task, and expressed how it had led to a series of productive actions later in the day:

"*When I got home, I took a rest, put on some music & cooked three different things - I don't know what came over me! I also cleaned up straight away - what is happening? Then I also scored my first job interview - what?! - which made me so happy!*".

She further attributes this success to the kit:

"*I have to say, today was a good day and if this project makes me feel like this, I'm all for it."* (Camilla, Journal entry, #15).

As a result of doing things that the users found beneficial, they started doing more things that they believed would enhance the same beneficial outcomes. We see this as a starting point for building habits that are more sustainable. In a design we would seek to motivate the user to take the first action, as we believe that through success experiences they would be able to build further upon these actions by themselves.

This explained 'first action' leads into what we have made a sub-code: reminders. A reminder serves the purpose of remembering the things that they enjoy doing, and that they might have forgotten about or stopped doing. It is about remembering the feeling they have when doing such things. This is also about being aware of things that will

benefit their well-being as well as keeping that as part of their daily ritual in order to create habits around things that will have a positive impact on their self-esteem. Our users experienced a plethora of positive reminders, in going for walks and recognising the positive impact it has or paying attention to the small things in life that make them happy. One of our users used one of the tasks, as a reminder to nurture and upkeep relationships, especially during the pandemic, where social contact is brief and minimal: "[The task] *also reminded me that if I don't want us to lose our friendship, then I should just write more often*" (Camilla, Journal entry, #12). This was especially beneficial to her, as she isn't used to writing or calling her friends, as it was out of her comfort zone to do so. After reaching out to a friend, she realised that it matters to her, to engage with her relationships, and wants to actively make that change going forward. Furthermore, through an activity task (Do something nice for yourself or for someone else), she was reminded that it is something that she values and recognises affects her self-esteem in a good way.

Reminders are often connected to specific actions that the users have positive experience with. The actions are very subjective to each person, and it can be very demanding to keep reminding oneself about which actions are more beneficial than others. Users tend to make a quick cost-benefit analysis on the action, which can demotivate them. For instance reading versus watching movies. Here, reading is more demanding, but the user finds it much more beneficial. In a design for building positive habits, reminders are very important, but complex as they shouldn't become an unnecessary stressor.

Design arguments

During our analysis we have identified several elements that could be a beneficial part of design for young women who want to build routines for self-appreciation. We have phrased these elements as design arguments. These arguments can be summed up as follows:

1. The design should consider the individuality of users, that people are unique and have different needs, because the solution should be adjustable to individual user needs.

2. The design should assist the users in readjusting their expectations, by acknowledging that guilt is part of changing behaviours, because guilt can cause ‘relapses’ in a change behaviour process.
3. The design should give users the ability to externalise their thoughts, as this will help them with reframing their reasoning in a more positive and rational manner.
4. The design should support and assist users in becoming familiar with and learning mindfulness practices because mindfulness can be difficult to engage with and have proven to be helpful in mitigating low self-esteem.
5. The design should allow users to restructure the way we compare ourselves, because comparison is inevitable.
6. The design should allow users to re-adjust self-perceptions by giving a space for users to recognise and affirm their value.
7. The design should distance the user from their smartphone, because the Social Media platforms are easily accessible here, and these may have a negative effect on self-esteem.
8. The design should allow for the complexities of building healthy and sustainable habits by focusing on creating compassionate encouragement and demanding patience from the user.

In the previous we have laid out our 8 design arguments that might prove helpful when designing for young women and assist them in practicing self-appreciation. In this context we have gained a stronger understanding of the design space by investigating our users' practices by applying the method of Cultural Probes. This method has been valuable in this context, as routines and habits are occurring over time. In order to investigate these, the researcher must be able to gather data in situ. Here, a large contribution from our work towards RtD has been the application of Cultural Probes and our ability to use the designed features of our probe kit to gain a deeper understanding of our users’ practices.

Further, our biggest contribution in regards to answering our research question is our design arguments. These work as a synthesis of our empirical work. Some of the most valuable learnings here are that when assisting young women in building self-appreciative routines, designers should incorporate subjectivity, reflection over actions and thought patterns, create space for being compassionate towards oneself, allowing the user to align expectations to the process of changing one’s behaviour, and attempt to work away from a screen-based application. In order to approach the design

arguments our workbook helped us concretise these in five design proposals. These allowed us to consider materials, spatial aspects and mediation of design arguments through use scenarios.

In the following we will explore how these design arguments could be approached through different design elements. We have created a workbook to present design proposals that integrate these elements in various ways.

Discussion

In this discussion, we will investigate how we have approached answering our research question and to what extent this can be regarded as successful. The discussion will focus particularly on two aspects of our research question: first, the question of working with digital artifacts, and second, working with long-term effects on behaviour.

Working towards digital artifacts

In our research question, we set out to understand how we can assist young women in practicing self-appreciation through a digital artifact. It can be discussed to which extent we have focused on the creation of a digital artifact. We have focused our research on assisting young women in practicing self-esteem, where we have allowed ourselves to linger on the context of self-appreciation and the various aspects that prove important to users. We have not had our primary focus on the development of a digital artifact, but rather explored its context, in order to have a solid foundation for further development of digital artifacts, beyond the scope of this thesis project. As we set out to start the process of working on this thesis, we were convinced that we would have time to develop a digital artifact that could also be tested, but soon we realised that we had unrealistic expectations for our project and decided to limit our expectations. We, therefore, focused on creating a cultural probe that could imitate design features that we would consider bringing into a digital artifact, based on findings from our design research. By doing this, we were able to simultaneously gather information about self-appreciation routines and information on the capabilities of specific (analogue) design features. If in turn, we had decided to prioritise the prototyping of digital artifacts in the design process, we would have had greater opportunity to user test and validate design features in our research context. This might have brought clarity in the extent to which we succeeded in addressing the first part of our research question.

We have later come to appreciate our decision to focus on understanding our context, rather than to focus on the development of an artifact earlier in the process, as we found that the context and user group we were working with was complex and we introduced problems that could, to some degree, be characterised as wicked

problems. This was particularly due to the complexities of self-esteem and subjectivity in ways of changing behaviour. This in turn led us to the creation of a design workbook, which would allow us to focus more widely on a solution area, rather than a single digital artifact. When working with wicked problems it can be more appropriate to work towards a solution area, as there can be many ways to approach the problem. Further, the problem might need more than one approach to cause an effect (Buchanan, 1992).

In our workbook, we have highlighted which design arguments each design proposal approaches. This was to clarify that it was not always possible to design for all possible design arguments (which was not our intention). Some design proposals moved further away from our initial goals and vision for our digital artifact. Especially in creating an artifact that moved away from interacting with a smartphone, which we had been particularly focused on in the initial part of our process, where we put forth a critique of the smartphone and social media by looking at the three forms of interaction design (Vallgårda, 2014). The most weighing argument in creating design proposals that would demand engaging with a smartphone was based on our user's feedback on their probe kit experiences. The feedback here focused on the wish for better mobility of the kit, where it would have offered better availability in a young person's life, which in these cases were dynamic and quite unstructured. For example, a young woman's routine often varies from day to day; university one day and work another – which in turn could also include sleeping at different locations throughout the week. This could make it difficult to engage with an analogue and semi-stationary probe kit. Therefore, we decided to create a digital smartphone application that would represent an iteration of the probe kit elements that seemed most agreeable with our users' needs, even though this would mean designing for a smartphone application. We see another strong argument for this in designing an artifact that is already familiar to the users and therefore would work with existing routines and habits, rather than working against these.

Other design proposals moved beyond our probe kit elements and were inspired more independently from our design arguments. The proposals that these arguments concern, are the three lamps and the virtual world for different personalities. Further, these proposals are inspired by our initial brainstorming sessions. We see that there are strengths and weaknesses in non-screen based digital artifacts, as well as screen

based. The screen based applications are better for adaptability for the users, where the non-screen based artifacts would help users gradually move away from their screens. We might even suggest a combined approach, where notifications would be moved from a screen based application to artifacts such as a lamp, as we have seen in our initial design critique, where notifications from social media on smartphones can be a stressful trigger for users. As these design proposals cannot be linked back to any elements from our probe kit, these do not have any foundation in user feedback, and we do see a limitation in these proposals because they would need further testing before they could be adopted into a design process.

Working towards long-term effects on behaviour

In the second aspect of our research question, we aimed to improve self-esteem in young women long-term. In the following, we will address the extent to which our design process and our findings are able to answer this aspect of our research question.

In order to tackle a wicked problem like self-esteem, we decided to take a Research through-Design (RtD) approach. We found through our research on self-esteem that it is complex and each person has a unique experience of self-esteem. We further wanted to understand how it manifests, what impacts it, implications of Social Media, and how to mitigate and/or understand feelings of low self-esteem. Through our research, we saw that it was an increasingly messy situation, with layers of complexity that complicates and challenges traditional design approaches. This was our initial motivation for using RtD as it is equipped to handle complex situations through its mixed-methods approach. RtD is further relevant to addressing our research question as it lends itself to investigating long-term practices.

Self-esteem is affected and impacted by a person’s lived experience, so in order to understand and design for the issue, lived experiences have to occur, which is only possible over a longer time scale. This is where cultural probes seemed of relevance as this works with principles of research through design. Cultural probes allowed us to understand and gain insights and knowledge of practices, through our users' lived experiences. Further, we were able to tailor our cultural probe to expose our users to

specific experiences that focused on self-esteem and the role it plays in a user's day to day life. Further, through our cultural probe, we were able to incorporate the principle of RtD that focuses on research of and for the future. We approached this by exposing our users to alternative so-called 'futures', i.e. presenting alternative realities, like applying mindfulness in their daily routines. Through their reflections in the journal, they were able to imagine these alternate realities as possible futures. As mentioned, RtD is often driven and defined by its mixed-methods approach and its integration of knowledge across disciplines (Zimmerman, 2010).

Our design process was dominated by our cultural probe, as the main data source and the primary way in which we gained an understanding into our context. It can be argued that we should have further incorporated additional methods, such as doing field-work in the home of our users that could be supported by in-depth semi-structured interviews. Cultural probes allow designers to gain a rich picture of a personal context. This is due to the fact that designers primarily exist in the onboarding and debriefing of a probe, meaning that the role of designer while a cultural probe is in play, is minimal. The expectation is that the users can document and reflect on their lived experiences, in the way that they feel most comfortable, thereby being 'user-sensitive' (Burrows et al., 2015). However, we could argue that having a larger role as a designer in our empirical work, through e.g. field-work, might have benefited our process and data collection. This could be exemplified in the ability to ask more structured and in-depth questions to our users in-situ about specific situations and scenarios that we would only be exposed to through being present in their context. Furthermore, using observations to gain tacit knowledge could have allowed a different perspective, contrary to just our user's perceived experiences. To counter the shortcomings of the cultural probe, we supported it with a questionnaire that aimed to gain an understanding of our context and the problem, as a whole. Essentially, attempting to legitimise our user's experiences as a representation of young women's lived experiences with self-esteem and self-appreciation.

We decided to collate our findings in the form of design arguments. The goal was to provide a design research theory in the form of a manifesto that should help designers create solutions for users practicing self-appreciation long-term. The design arguments were created based on insights that we gathered in our probe kit and

through the workshop. In those, we attempted to create a space for self-appreciation. This can be exemplified in arguments that promote patience and compassion. Further, the arguments are in their essence applicable over a longer time scale. In order for us to assess whether our arguments actually live up to the goal of improving self-esteem long-term, it would require further user testing of our design proposals, of which we have been unable to carry out in our set time frame. As explained earlier in our report, we discussed evaluation within the field of HCI. In this instance, we explained it in relation to the value of our cultural probe and our workshop. We see relevance and necessity in further using these principles of evaluation in the assessment of our design proposals. Although our time frame has limited our design outcome to be in the form of design arguments supported by our design proposals, the natural continuation of our design process would require an assessment of these design proposals in their imagined context. First through a long-term contextual assessment, would we be able to value whether our design arguments do in fact help when addressing self-esteem and practicing self-appreciation with young women long-term.

Conclusion

To reiterate, our research question was focused on the creation of a digital artefact that could assist young women in practicing self-appreciation and, ideally, improve their self-esteem and happiness long-term.

In order to answer this research question we set out to investigate our research context by doing design research and research through design. Here, we created a cultural probe consisting of several tasks that were founded in our initial research. These tasks were chosen in order to test what methods and techniques would be most beneficial when assisting young women in practicing self-appreciation. We also implicitly tested possible design features that could later be applied in design proposals. We then evaluated our probe kit results through a participatory workshop with all of our users. Thereafter, we coded and analysed our empirical data that led to our 8 design arguments. From these arguments, we developed a design workbook containing various design proposals that help substantiate the design arguments as a legitimate design theory. We consider our biggest contribution to the context in which we have done design research to be our design arguments and our design workbook. Here, our design arguments are specifically addressing how to design for this context and our design proposals help in showing how these design arguments come to life.

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